

# Package ‘mathmodels’

July 28, 2026

**Type** Package

**Title** Mathematical Modeling Algorithms

**Version** 0.0.11

**Description** Implements algorithms for mathematical modeling across evaluation, prediction, and differential equation systems. It supports indicator preprocessing, weighting, TOPSIS, fuzzy comprehensive evaluation, grey relational analysis, rank sum ratio, data envelopment analysis (DEA), regional economics, inequality measures, grey prediction, Markov prediction, and time series modeling. It also provides ordinary differential equation solvers for compartmental epidemic models together with visualization and summary metrics.

**License** AGPL (>= 3)

**URL** <https://github.com/zhjx19/mathmodels>, <https://zhjx19.github.io/mathmodels/>

**BugReports** <https://github.com/zhjx19/mathmodels/issues>

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dplyr (>= 1.0.0),  
forecast,  
ggplot2 (>= 3.5.0),  
MASS,  
purrr,  
readxl,  
rlang,  
stats,  
stringr,  
tibble,  
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patchwork,  
car,  
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AHP

*AHP: Analytic Hierarchy Process***Description**

AHP is a multi-criteria decision analysis method developed by Saaty, which can also be used to determine indicator weights.

**Usage**

AHP(A)

**Arguments**

A a numeric matrix, i.e. pairwise comparison matrix

**Value**

a list object that contains: w (Weight vector), CR (Consistency ratio), Lmax (Maximum eigenvalue), CI (Consistency index)

**Examples**

```
A = matrix(c(1, 1/2, 4, 3, 3,
             2, 1, 7, 5, 5,
             1/4, 1/7, 1, 1/2, 1/3,
             1/3, 1/5, 2, 1, 1,
             1/3, 1/5, 3, 1, 1), byrow = TRUE, nrow = 5)
AHP(A)
```

as\_ts\_df

*Convert Common Inputs to ts\_df***Description**

Converts a ts object, numeric vector, explicit data-frame columns, or an existing ts\_df to the package's lightweight time-series structure.

**Usage**

```
as_ts_df(x, time = NULL, value = NULL, frequency = NULL, start = NULL)
```

**Arguments**

|             |                                               |
|-------------|-----------------------------------------------|
| x           | A ts, numeric vector, data frame, or ts_df.   |
| time, value | Required column names when x is a data frame. |
| frequency   | Optional frequency metadata.                  |
| start       | Optional start metadata.                      |

**Value**

A ts\_df object.

**Examples**

```
as_ts_df(ts(1:4, frequency = 4))
as_ts_df(c(3, 5, 8))
as_ts_df(data.frame(month = 1:3, demand = c(10, 12, 15)), time = "month", value = "demand")
```

---

`combine_preds`*Combine Multiple Prediction Results*

---

**Description**

Combines multiple prediction results (e.g., from grey prediction, time series, or machine learning models) into a single prediction using a similarity-based weighting approach, improving prediction accuracy.

**Usage**

```
combine_preds(x)
```

**Arguments**

`x` Numeric vector, prediction results to be combined (length  $\geq 2$ ).

**Details**

The function combines prediction results by constructing a similarity matrix based on cosine transformation of pairwise differences. Weights are derived from the principal eigenvector of the similarity matrix, ensuring predictions closer to each other have higher influence. For two predictions, equal weights (0.5, 0.5) are used. If all predictions are identical, equal weights are assigned. Compatible with the `mathmodels` package for enhancing prediction models, including grey prediction, time series, or ensemble machine learning.

**Value**

A list with two elements:

- `a`: Numeric, the combined prediction value.
- `w`: Numeric vector, weights for each prediction in `x`, summing to 1.

**Examples**

```
# Example: Combine three prediction results
preds = c(100, 102, 98) # E.g., from grey prediction, ARIMA, or ML models
combine_preds(preds)
```

---

`combine_weights`*Combine Subjective and Objective Weights*

---

**Description**

Combines subjective and objective weights using linear, multiplicative, or game theory-based methods (geometric mean or linear system).

**Usage**

```
combine_weights(w_subj, w_obj, type = "linear", alpha = 0.5)
```

**Arguments**

|        |                                                                                                               |
|--------|---------------------------------------------------------------------------------------------------------------|
| w_subj | Numeric vector of subjective weights.                                                                         |
| w_obj  | Numeric vector of objective weights.                                                                          |
| type   | Character string specifying the combination method: "linear", "multiplicative", "game", or "game_linear".     |
| alpha  | Numeric value between 0 and 1, used only for the linear method to weight subjective weights. Defaults to 0.5. |

**Details**

The function supports four methods:

- Linear: Combines weights as  $\alpha * w_{\text{subj}} + (1 - \alpha) * w_{\text{obj}}$ .
- Multiplicative: Combines weights as  $w_{\text{subj}} * w_{\text{obj}}$ , requiring positive weights.
- Game: Uses the geometric mean ( $\sqrt{w_{\text{subj}} * w_{\text{obj}}}$ ) to balance weights.
- Game\_linear: Uses a game-theoretic approach by solving a linear system based on the cross-product of weights.

**Value**

A numeric vector of combined weights, normalized to sum to 1.

**Examples**

```
w_subj = c(0.4, 0.3, 0.2, 0.1)
w_obj = c(0.25, 0.2, 0.3, 0.25)
combine_weights(w_subj, w_obj, type = "linear", alpha = 0.6)
combine_weights(w_subj, w_obj, type = "multiplicative")
combine_weights(w_subj, w_obj, type = "game")
combine_weights(w_subj, w_obj, type = "game_linear")
```

---

complete\_ts\_df

---

*Complete Missing Time Points*


---

**Description**

Adds missing time points on a regular sequence and leaves their values as NA.

**Usage**

```
complete_ts_df(x, by)
```

**Arguments**

|    |                            |
|----|----------------------------|
| x  | A ts_df.                   |
| by | Step size passed to seq(). |

**Value**

A completed ts\_df.

**Examples**

```
x = ts_df(data.frame(time = c(1, 2, 4), value = c(10, 12, 18)))
complete_ts_df(x, by = 1)
```

compute\_mf

*Compute fuzzy membership vector and return corresponding membership functions.*

**Description**

compute\_mf transforms a single indicator value into a fuzzy membership vector, where each element represents the degree of membership to a specific evaluation level. compute\_mf\_funs returns the list of membership functions for visualization purposes.

**Usage**

```
compute_mf_funs(thresholds)
```

```
compute_mf(x, thresholds)
```

**Arguments**

|            |                                                                                                                 |
|------------|-----------------------------------------------------------------------------------------------------------------|
| thresholds | A numeric vector containing at least two threshold values that define the boundaries between evaluation levels. |
| x          | A numeric scalar representing the value of an indicator.                                                        |

**Value**

A list with two elements:

**mv** A numeric vector, membership degrees to each level.

**mfs** A list of functions, one per level, for plotting membership functions.

**Examples**

```
# Example: SO2 concentration = 0.07, thresholds = c(0.05, 0.15, 0.25, 0.5)
th = c(0.05, 0.15, 0.25, 0.5)
compute_mf(0.07, th)

## Not run:
mfs = compute_mf_funs(th)
plots = lapply(mfs, \(x) plot_mf(x, xlim = c(0, 0.6)))
gridExtra::grid.arrange(grobs = plots, nrow = 2)

## End(Not run)
```

---

|               |                             |
|---------------|-----------------------------|
| critic_weight | <i>CRITIC Weight Method</i> |
|---------------|-----------------------------|

---

**Description**

Computes objective weights of indicators and scores of samples using the CRITIC method. The method considers both the contrast intensity (e.g., standard deviation or entropy) and conflict among indicators (based on correlation) to determine indicator importance. This version supports different methods for contrast intensity and correlation types.

**Usage**

```
critic_weight(  
  X,  
  index = NULL,  
  method = "std",  
  cor_method = "pearson",  
  epsilon = 0.002  
)
```

**Arguments**

|            |                                                                                                                                                                                                                                                                                                                                   |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| X          | A numeric data frame or matrix where rows represent samples (observations) and columns represent indicators (variables).                                                                                                                                                                                                          |
| index      | A character vector indicating the direction of each indicator: Use "+" for positive indicators (higher is better), "-" for negative indicators (lower is better), and NA for already normalized indicators (no rescaling will be applied).<br><br>If `index = NULL` (default), all indicators are treated as `NA`, meaning no nor |
| method     | Character scalar; specifies the method used to compute contrast intensity. Options: "std" (standard deviation, default), or "entropy" (based on information redundancy).                                                                                                                                                          |
| cor_method | Character scalar; specifies the method for computing correlations. Options: "pearson" (default), "spearman", or "kendall".                                                                                                                                                                                                        |
| epsilon    | A small constant used to replace exact 0s and 1s in the data to prevent log(0) errors. Default is 0.002. Only used when method = "entropy".                                                                                                                                                                                       |

**Value**

|                    |                                                                |
|--------------------|----------------------------------------------------------------|
| A list containing: |                                                                |
| w                  | Numeric vector of weights for each indicator.                  |
| s                  | Numeric vector of scores for each sample (row), scaled by 100. |

**Examples**

```
# Example: Using CRITIC method on a simple dataset  
X = data.frame(  
  x1 = c(3, 5, 2, 7),  
  x2 = c(10, 20, 15, 25)  
)
```



```

index = c("+", "-")
critic_weight(X, index)
critic_weight(X, index, method = "entropy")

```

curve\_fit

*Linearizable Curve Fit*

## Description

Fits common empirical curves via variable transformation and `lm()`. Supported types: exponential, power-law, logarithmic, and hyperbolic.

## Usage

```
curve_fit(x, y, type = c("exp", "power", "log", "hyperbolic"))
```

## Arguments

|                   |                                                        |
|-------------------|--------------------------------------------------------|
| <code>x</code>    | Numeric vector of observed predictor values.           |
| <code>y</code>    | Numeric vector of observed response values.            |
| <code>type</code> | Character, one of "exp", "power", "log", "hyperbolic". |

## Details

Each curve is transformed to a linear form:

- "exp": fits  $\log(y) \sim x$ , back-transforms to  $y = a * \exp(b * x)$
- "power": fits  $\log(y) \sim \log(x)$ , back-transforms to  $y = a * x^b$
- "log": fits  $y \sim \log(x)$ , gives  $y = a + b * \log(x)$
- "hyperbolic": fits  $y \sim 1/x$ , gives  $y = a + b / x$

Fit statistics are computed on the original scale.

**Note on log-transform bias:** For "exp" and "power" types, fitted values are back-transformed directly ( $a * \exp(b * x)$  or  $a * x^b$ ), which yields the geometric-mean prediction. As a result, predicted values on the original scale are systematically biased downward by a factor of approximately  $\exp(\sigma^2 / 2)$ . This does not affect the regression coefficients (unbiased on the log scale) or the goodness-of-fit statistics (computed on the original scale).

**Note on coefficients:** For "exp" and "power" types, the coefficient table reports estimates on the log-transformed scale (the parameterization of the underlying `lm()` model). The formula string shows the back-transformed parameters on the original scale.

## Value

A named list, see [poly\\_fit\(\)](#) for details.

## Examples

```

set.seed(42)
x = 1:20
y = 2 * exp(0.15 * x) * rlnorm(20, sdlog = 0.1)
curve_fit(x, y, type = "exp")

```

---

 cv\_weight

*Coefficient of Variation Weighting*


---

### Description

Computes weights for indicators using the Coefficient of Variation (CV) method. Weights are derived by normalizing the CV (standard deviation divided by mean) for each indicator.

### Usage

```
cv_weight(X)
```

### Arguments

X                      Numeric matrix or data frame with positive indicator data.

### Details

The `cv_weight` function calculates weights using the CV method. For each column in data, the CV is computed as the standard deviation divided by the mean. Weights are obtained by normalizing the CVs to sum to 1. This lightweight implementation uses base R and assumes all columns are numeric indicators.

### Value

Numeric vector of weights for the indicators, summing to 1.

### Examples

```
X = data.frame(x1 = c(10, 20, 15), x2 = c(5, 10, 8))
cv_weight(X)
```

---

 DEA

*DEA efficiency analysis*


---

### Description

DEA efficiency analysis

### Usage

```
basic_DEA(
  data,
  inputs,
  outputs,
  ud_outputs = NULL,
  orientation = "io",
  rts = "vrs"
)
```

```

super_DEA(data, inputs, outputs, orientation = "io", rts = "vrs")

basic_SBM(
  data,
  inputs,
  outputs,
  ud_outputs = NULL,
  orientation = "io",
  rts = "vrs"
)

super_SBM(data, inputs, outputs, orientation = "io", rts = "vrs")

malmquist(
  data,
  period,
  inputs,
  outputs,
  orientation = "oo",
  rts = "vrs",
  type1 = "glob",
  type2 = "rd"
)

```

### Arguments

|             |                                                                                                                                                                                                                            |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| data        | A data frame. 1st column = DMU names.                                                                                                                                                                                      |
| inputs      | Column indices/names of input variables.                                                                                                                                                                                   |
| outputs     | Column indices/names of output variables.                                                                                                                                                                                  |
| ud_outputs  | Column indices (within outputs) of undesirable outputs, or NULL.                                                                                                                                                           |
| orientation | "io" (input-oriented, default) or "oo" (output-oriented). All DEA functions return Farrell efficiency (0~1), where 1 = efficient. For output orientation: value = $1/\phi$ , where $\phi \geq 1$ is the Shephard distance. |
| rts         | "vrs" (variable returns to scale, default) or "crs" (constant).                                                                                                                                                            |
| period      | Column name or index identifying the time period variable.                                                                                                                                                                 |
| type1       | "cont" (contemporaneous), "seq" (sequential), or "glob" (global, default).                                                                                                                                                 |
| type2       | "fgnz" (Färe-Grosskopf-Norris-Zhang) or "rd" (Ray-Desli, default).                                                                                                                                                         |

### Value

A list with components: efficiencies, lambdas, slacks, targets, returns, model, orientation, dmu.

### Functions

- `basic_DEA()`: Standard radial DEA (CCR/BCC)
- `super_DEA()`: Super-efficiency DEA (self excluded, radial only)
- `basic_SBM()`: Standard Slacks-Based Measure (SBM, Tone 2001)
- `super_SBM()`: Super-efficiency SBM (self excluded, no undesirable outputs)
- `malmquist()`: Malmquist productivity index

## Examples

```
df = data.frame(
  DMU = paste0("DMU", 1:7),
  x1 = c(20, 60, 40, 60, 70, 30, 50),
  x2 = c(151, 200, 120, 170, 250, 210, 90),
  y1 = c(100, 210, 150, 240, 220, 80, 200)
)
basic_DEA(df, inputs = 2:3, outputs = 4, rts = "crs")
basic_DEA(df, inputs = 2:3, outputs = 4, rts = "vrs")
df = data.frame(
  DMU = paste0("DMU", 1:7),
  x1 = c(20, 60, 40, 60, 70, 30, 50),
  x2 = c(151, 200, 120, 170, 250, 210, 90),
  y1 = c(100, 210, 150, 240, 220, 80, 200)
)
super_DEA(df, inputs = 2:3, outputs = 4, rts = "crs")
df = data.frame(
  DMU = paste0("DMU", 1:7),
  x1 = c(20, 60, 40, 60, 70, 30, 50),
  x2 = c(151, 200, 120, 170, 250, 210, 90),
  y1 = c(100, 210, 150, 240, 220, 80, 200)
)
basic_SBM(df, inputs = 2:3, outputs = 4, rts = "crs")
df = data.frame(
  DMU = paste0("DMU", 1:7),
  x1 = c(20, 60, 40, 60, 70, 30, 50),
  x2 = c(151, 200, 120, 170, 250, 210, 90),
  y1 = c(100, 210, 150, 240, 220, 80, 200)
)
super_SBM(df, inputs = 2:3, outputs = 4, rts = "crs")
panel = data.frame(
  DMU = rep(paste0("DMU", 1:5), 3),
  Period = rep(1:3, each = 5),
  x1 = c(10, 20, 15, 25, 30, 12, 22, 17, 27, 32, 14, 24, 19, 29, 34),
  y1 = c(100, 150, 120, 180, 200, 110, 160, 130, 190, 210, 120, 170, 140, 200, 220)
)
malmquist(panel, period = "Period", inputs = 3, outputs = 4,
  rts = "crs", type1 = "cont", type2 = "fgnz")
```

---

defuzzify

*Defuzzification Methods for Fuzzy Comprehensive Evaluation*


---

## Description

Implements defuzzification methods for fuzzy evaluation vectors, including weighted average and maximum membership methods.

## Usage

```
defuzzify(mu, scores, method = "weighted_average")
```

**Arguments**

|        |                                                                                                                                   |
|--------|-----------------------------------------------------------------------------------------------------------------------------------|
| mu     | Numeric vector, membership degrees for evaluation levels, in $[0, 1]$ .                                                           |
| scores | Numeric vector, scores corresponding to each evaluation level (e.g., c(100, 80, 60, 40) for "Excellent", "Good", "Fair", "Poor"). |
| method | Character, defuzzification method: "weighted_average", "max_membership", "centroid".                                              |

**Value**

Numeric, defuzzified output value.

**Examples**

```
# Example: Defuzzify fuzzy evaluation vectors for three schemes
mu = c(0.318, 0.351, 0.203, 0.128)
scores = c(30, 60, 75, 90) # Scores for "Poor", "Fair", "Good", "Excellent"
defuzzify(mu, scores, method = "weighted_average")
defuzzify(mu, scores, method = "max_membership")
defuzzify(mu, scores, method = "centroid")
```

---

drop\_na\_ts\_df

*Drop Missing Values from a ts\_df*


---

**Description**

Drop Missing Values from a ts\_df

**Usage**

```
drop_na_ts_df(x)
```

**Arguments**

|   |          |
|---|----------|
| x | A ts_df. |
|---|----------|

**Value**

A ts\_df with rows whose value is NA removed.

**Examples**

```
x = ts_df(data.frame(time = 1:4, value = c(10, NA, 12, 13)))
drop_na_ts_df(x)
```

---

|                |                              |
|----------------|------------------------------|
| entropy_weight | <i>Entropy Weight Method</i> |
|----------------|------------------------------|

---

## Description

Computes the weights of indicators and scores of samples based on the entropy method. This method objectively determines the importance of each indicator according to the amount of information it contains.

## Usage

```
entropy_weight(X, index = NULL, epsilon = 0.002)
```

## Arguments

|                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>X</code>       | A numeric data frame or matrix where rows represent samples (observations) and columns represent indicators (variables).                                                                                                                                                                                                                                                                                                                                                                                    |
| <code>index</code>   | A character vector indicating the direction of each indicator. Use "+" for positive indicators (higher is better), "-" for negative indicators (lower is better), and NA for already normalized indicators (no rescaling will be applied, but minor adjustments will still be made to avoid log(0) errors). If <code>index = NULL</code> (default), all indicators are treated as NA, meaning no normalization or rescaling is performed, but a small adjustment is still applied to prevent log(0) errors. |
| <code>epsilon</code> | A small constant used to replace exact 0s and 1s in the data to prevent log(0) errors. Default is 0.002.                                                                                                                                                                                                                                                                                                                                                                                                    |

## Value

A list containing:

|                |                                                                |
|----------------|----------------------------------------------------------------|
| <code>w</code> | Numeric vector of weights for each indicator.                  |
| <code>s</code> | Numeric vector of scores for each sample (row), scaled by 100. |

## Examples

```
X = data.frame(
  x1 = c(3, 5, 2, 7),
  x2 = c(10, 20, 15, 25)
)
index = c("+", "-")
entropy_weight(X, index)
```

epi\_metrics

*Extract key epidemic metrics from ODE simulation output***Description**

This function computes core epidemiological summary statistics from deterministic compartmental epidemic models (e.g., SIR, SIS, SEIR).

**Usage**

```
epi_metrics(data, beta, gamma, N = NULL)
```

**Arguments**

|       |                                                                                                                                                                                                                                     |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| data  | A data frame returned by <code>model_*()</code> functions. Must contain a column <code>time</code> , and at least <code>S</code> and <code>I</code> . Additional compartments such as <code>E</code> or <code>R</code> are allowed. |
| beta  | Transmission rate (numeric).                                                                                                                                                                                                        |
| gamma | Recovery / removal rate (numeric).                                                                                                                                                                                                  |
| N     | Total population (numeric). If <code>NULL</code> (default), it is computed as the sum of all compartment values at the first time point.                                                                                            |

**Details**

The input data frame is expected to be the output of `model_*()` functions, containing a `time` column and one or more compartment columns (e.g., `S`, `I`, `R`, `E`). All compartment values are assumed to be in absolute population counts, and the total population is assumed to be conserved unless otherwise specified.

**Value**

A named list with:

**R0** Basic reproduction number.

**peak\_infection** Maximum number of infectious individuals.

**peak\_time** Time at which infectious peak occurs.

**attack\_rate** Proportion of population that transitioned out of `S`; defined only for models containing an `R` compartment (e.g., `SIR/SEIR`).

**Examples**

```
sir = model_sir(
  init   = c(S = 990, I = 10),
  params = c(beta = 0.002, gamma = 0.1),
  times  = seq(0, 50, by = 0.1)
)
epi_metrics(sir, beta = 0.002, gamma = 0.1)
```

fuzzy\_eval

*Fuzzy Comprehensive Evaluation***Description**

Performs fuzzy comprehensive evaluation using different fuzzy composition operators to combine factor weights with a fuzzy evaluation matrix. Suitable for multi-criteria decision analysis with weights from methods like AHP, entropy, CRITIC, CV, or PCA.

**Usage**

```
fuzzy_eval(w, R, type)
```

**Arguments**

|      |                                                                                                                                                                                                                                                                                                                                                                           |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| w    | Numeric vector, factor weights (e.g., from <code>combine_weights</code> ).                                                                                                                                                                                                                                                                                                |
| R    | Numeric matrix, fuzzy evaluation matrix with columns as factors and rows as evaluation grades. Values should be in $[0, 1]$ .                                                                                                                                                                                                                                             |
| type | Integer or character (1-5), specifying the fuzzy composition operator: <ul style="list-style-type: none"> <li>• 1: Min-max (main factor decisive).</li> <li>• 2: Product-max (main factor prominent).</li> <li>• 3: Weighted sum (additive average).</li> <li>• 4: Bounded sum of mins (min-sum bounded).</li> <li>• 5: Normalized min-sum (balanced average).</li> </ul> |

**Details**

The function computes a fuzzy comprehensive evaluation vector  $B$  based on the weight vector  $w$  and fuzzy evaluation matrix  $R$ . Five composition operators are supported:

- Type 1 (min-max):  $\max(\min(w, R[j, ]))$ , emphasizes the main factor.
- Type 2 (product-max):  $\max(w * R[j, ])$ , highlights the main factor.
- Type 3 (weighted sum):  $\text{sum}(w * R[j, ])$ , additive average.
- Type 4 (bounded sum):  $\min(1, \text{sum}(\min(w, R[j, ])))$ , bounds the sum of mins.
- Type 5 (normalized min-sum):  $\text{sum}(\min(w, R[j, ] / \text{sum}(R[j, ])))$ , balanced average.

The output  $B$  is normalized to sum to 1. If the sum is zero, an error is thrown. Uses base R for lightweight implementation.

**Value**

A numeric vector of normalized comprehensive evaluation results, summing to 1.



**Examples**

```
w = c(0.3, 0.3, 0.3, 0.1) # weights (e.g., from AHP or entropy)

# fuzzy evaluation matrix (3 grades for 4 factors)
R = matrix(c(0.8, 0.7, 0.6, 0.7,
             0.1, 0.2, 0.2, 0.1,
             0.1, 0.1, 0.2, 0.2), nrow = 3, byrow = TRUE)
# Apply fuzzy comprehensive evaluation
fuzzy_eval(w, R, type = 3) # Weighted sum
```

GM11\_markov

*Grey-Markov Prediction Model***Description**

Fits a GM(1,1) model and then uses a Markov chain on relative error states to correct both historical fitted values and future predictions.

**Usage**

```
GM11_markov(X, n_ahead = 3, breaks = c(-0.1, -0.05, -0.02, 0, 0.02, 0.05, 0.1))
```

**Arguments**

|         |                                                                                                                           |
|---------|---------------------------------------------------------------------------------------------------------------------------|
| X       | Numeric vector of original time series data.                                                                              |
| n_ahead | Number of future periods to predict. Must be a positive integer.                                                          |
| breaks  | Numeric vector of boundaries for classifying relative errors into states. -Inf and Inf are prepended/appended internally. |

**Details**

The function first fits a GM(1,1) model to the original series, classifies relative errors into states using breaks, builds a Markov chain on these error states, and finally adjusts the GM(1,1) fitted and predicted values.

**Value**

A data frame with columns:

**Period** Time period labels, such as T1, T2, ..., T+n.

**Raw** Original values; future periods are NA.

**GM11\_fitted** GM(1,1) fitted or predicted values.

**err\_state** Relative error state for historical periods, and predicted state for future periods.

**adj\_eff** Adjustment effect, defined as the midpoint of the state interval.

**Markov\_adj** Markov-chain-adjusted fitted or predicted values.

**Examples**

```
X = c(174, 179, 183, 189, 207, 234, 220.5, 256, 270, 285)
GM11_markov(X, n_ahead = 3)
```

**Description**

A collection of functions for performing grey relational analysis, including calculation of grey correlation degree and evaluation based on grey correlation. These functions are designed for decision-making and data analysis by measuring the relational degree between sequences.

**Usage**

```
grey_corr(ref, cmp, rho = 0.5, w = NULL)
```

```
grey_corr_topsis(X, w, index = NULL, rho = 0.5)
```

**Arguments**

|       |                                                                                                                                                                                                                                                                                                                                         |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ref   | Numeric vector, the reference sequence for <code>grey_corr</code> .                                                                                                                                                                                                                                                                     |
| cmp   | Numeric matrix or data frame, the comparison sequences for <code>grey_corr</code> .                                                                                                                                                                                                                                                     |
| rho   | Numeric scalar, the distinguishing coefficient (default = 0.5).                                                                                                                                                                                                                                                                         |
| w     | Numeric vector, weights for weighted correlation (default = equal weights).                                                                                                                                                                                                                                                             |
| X     | Numeric matrix or data frame, the decision matrix for <code>grey_corr_topsis</code> .                                                                                                                                                                                                                                                   |
| index | Character vector indicating indicator direction: Use "+" for positive indicators (higher is better), "-" for negative indicators (lower is better), and NA for already rescaled indicators (no rescaling will be applied). If <code>index = NULL</code> (default), all indicators are treated as NA, meaning no rescaling is performed. |

**Details**

These functions implement grey relational analysis for evaluating relationships between sequences or decision alternatives:

**grey\_corr** Computes the grey correlation degree between a reference sequence (`ref`) and comparison sequences (`cmp`) using the distinguishing coefficient (`rho`) and optional weights (`w`).

**grey\_corr\_topsis** Evaluates a decision matrix (`X`) by normalizing it, applying weights (`w`), computing grey correlation with the ideal sequence. Direction of indicators can be specified via `index`.

**Value**

**grey\_corr** Returns a numeric vector of grey correlation degrees for each comparison sequence.

**grey\_corr\_topsis** Returns a numeric vector of relative closeness (grey correlation degrees).

**Examples**

```
# Grey correlation degree
ref = c(0.9, 0.8, 0.7)
cmp = data.frame(
  x1 = c(0.9, 0.7, 0.8),
  x2 = c(0.8, 0.9, 0.7),
```

```

    x3 = c(0.7, 0.8, 0.9)
  )
  grey_corr(ref, cmp, rho = 0.5)

  # Grey correlation evaluation
  X = data.frame(x1 = c(8, 7, 6), x2 = c(150, 180, 200), x3 = c(60, 80, 100))
  w = c(0.3, 0.4, 0.3)
  idx = c("+", "+", "+")
  grey_corr_topsis(X, w, idx, rho = 0.5)

```

grey\_models

*Grey Prediction Models***Description**

Implements grey prediction models for time series forecasting: GM11 applies the GM(1,1) model with level ratio test. GM1N applies the GM(1,N) model with multiple related factors. DGM21 applies the DGM(2,1) model for second-order dynamics. verhulst applies the Verhulst model for logistic growth.

**Usage**

```

GM11(X)

GM1N(dat, new_data = NULL)

DGM21(X)

verhulst(X)

```

**Arguments**

|          |                                                                                                   |
|----------|---------------------------------------------------------------------------------------------------|
| X        | For GM11, DGM21, verhulst: Numeric vector of original time series data.                           |
| dat      | For GM1N: Data frame or matrix, last column is characteristic series, others are related factors. |
| new_data | For GM1N: Optional future values of related factors (1 row, m-1 columns).                         |

**Value**

For GM11: List with fitted values (`fitted`), next prediction (`pnext`), prediction function (`f`), matrix (`mat`), parameters (`u`), level ratios (`lambda`), and range (`rng`). For GM1N: List with fitted values (`fitted`), posterior variance ratio (`C`), small error probability (`P`), and prediction function (`f`). For DGM21, verhulst: List with fitted values (`fitted`), next prediction (`pnext`), prediction function (`f`), matrix (`mat`), and parameters (`u`).

**Examples**

```

# Sample time series for GM11, DGM21, Verhulst
x = c(100, 120, 145, 175, 210)

# GM11

```

```

result = GM11(x)
result$fitted      # Fitted values
result$pnext       # Next prediction
result$f(6:8)      # Predict next 3 periods

# DGM21
x = c(2.874,3.278,3.39,3.679,3.77,3.8)
result = DGM21(x)
result$fitted      # Fitted values
result$pnext       # Next prediction
result$f(6:8)      # Predict next 3 periods

# Verhulst
x = c(4.93,2.33,3.87,4.35,6.63,7.15,5.37,6.39,7.81,8.35)
result = verhulst(x)
result$fitted      # Fitted values
result$pnext       # Next prediction
result$f(6:8)      # Predict next 3 periods

# Sample data for GM1N
data = data.frame(
  factor1 = c(50, 55, 60, 65, 70),
  factor2 = c(20, 22, 25, 28, 30),
  output = c(100, 120, 145, 175, 210)
)
result = GM1N(data)
result$fitted

```

---

growth\_fit

Nonlinear Growth Curve Fit

---

## Description

Fits nonlinear growth, saturation, and sigmoid curves via `minpack.lm::nlsLM()` with automatic starting value generation.

## Usage

```
growth_fit(x, y, type = c("logistic", "gomPERTZ", "saturation", "mm"))
```

## Arguments

|      |                                                                  |
|------|------------------------------------------------------------------|
| x    | Numeric vector of observed predictor values.                     |
| y    | Numeric vector of observed response values.                      |
| type | Character, one of "logistic", "gomPERTZ", "saturation", or "mm". |

## Details

Models and starting value strategies:

**Logistic:**  $y = L / (1 + \exp(-k * (x - x_0)))$

- $L_0 = \max(y) * 1.05$

- $\text{lm}(\log((L_0 - y) / y) \sim x)$  for  $k$ ,  $x_0$  initials

**Gompertz:**  $y = L * \exp(-\exp(-k * (x - x_0)))$

- $L_0 = \max(y) * 1.05$
- $\text{lm}(\log(-\log(y / L_0)) \sim x)$  for  $k$ ,  $x_0$  initials

**Saturation** (exponential approach):  $y = a * (1 - \exp(-b * x))$

- $a_0 = \max(y) * 1.1$
- $\text{lm}(\log(a_0 - y) \sim x)$  for  $b$  initial

**Michaelis-Menten:**  $y = a * x / (b + x)$

- $a_0 = \max(y) * 1.1$
- Lineweaver-Burk  $\text{lm}(1/y \sim 1/x)$  for  $a$ ,  $b$  initials

Falls back to heuristic defaults if linearized estimation fails.

### Value

A named list, see [poly\\_fit\(\)](#) for details. `model` contains the `nls` object.

### Examples

```
set.seed(42)
x = seq(0, 10, length.out = 30)
y = 100 / (1 + exp(-1.5 * (x - 5))) + rnorm(30, sd = 2)
growth_fit(x, y, type = "logistic")
```

---

impute\_ts\_df

---

Complete and Interpolate a ts\_df

---

### Description

Completes missing time points and fills internal missing values by linear or spline interpolation. Endpoint values must already be observed.

### Usage

```
impute_ts_df(x, by, method = c("linear", "spline"))
```

### Arguments

|                     |                                             |
|---------------------|---------------------------------------------|
| <code>x</code>      | A <code>ts_df</code> .                      |
| <code>by</code>     | Step size passed to <code>seq()</code> .    |
| <code>method</code> | Interpolation method: "linear" or "spline". |

### Value

An interpolated `ts_df`.

**Examples**

```
x = ts_df(data.frame(time = c(1, 3, 5), value = c(10, 20, 30)))
impute_ts_df(x, by = 1, method = "linear")
```

inequality

*Inequality Indices***Description**

Computes inequality indices for individual or grouped data: `gini0` calculates the Gini coefficient for individual sample data. `gini` calculates the Gini coefficient for grouped data using income and population shares. `theil0` calculates the Theil index for individual sample data. `theil` calculates the Theil index for grouped average data. `theil0_g` calculates the Theil index and decomposition for grouped sample data. `theil_g` calculates the Theil index and decomposition for grouped average data. `theil_g2_cross` calculates the Theil index and decomposition for two-level cross-grouped average data. `theil_g2_nest` calculates the Theil index and decomposition for two-level nested grouped average data.

For `theil`, `theil_g`, `theil_g2_cross`, `theil_g2_nest`: Name of population variable (character).

**Usage**

```
gini0(y)

gini(y, pop)

theil0(y)

theil(y, pop)

theil0_g(data, group, y)

theil_g(data, group, y, pop)

theil_g2_cross(data, group1, group2, y, pop)

theil_g2_nest(data, group1, group2, y, pop)
```

**Arguments**

|                     |                                                                                                                                                                                                                                                           |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>y</code>      | For <code>gini0</code> , <code>gini</code> , <code>theil0</code> : Numeric vector of individual incomes.                                                                                                                                                  |
| <code>pop</code>    | For <code>gini</code> : Numeric vector of group populations or population shares. For <code>theil</code> , <code>theil0_g</code> , <code>theil_g</code> , <code>theil_g2_cross</code> , <code>theil_g2_nest</code> : Name of income variable (character). |
| <code>data</code>   | For <code>theil0_g</code> , <code>theil_g</code> , <code>theil_g2_cross</code> , <code>theil_g2_nest</code> : Data frame containing variables.                                                                                                            |
| <code>group</code>  | For <code>theil0_g</code> , <code>theil_g</code> : Name of grouping variable (e.g., province).                                                                                                                                                            |
| <code>group1</code> | For <code>theil_g2_cross</code> , <code>theil_g2_nest</code> : Name of first grouping variable (e.g., region or province).                                                                                                                                |
| <code>group2</code> | For <code>theil_g2_cross</code> , <code>theil_g2_nest</code> : Name of second grouping variable (e.g., type or city).                                                                                                                                     |

**Value**

For gini0, gini: Numeric Gini coefficient (0 to 1). For theil0, theil: Numeric Theil index. For theil0\_g, theil\_g, theil\_g2\_cross, theil\_g2\_nest: List with two vectors:

- theil: theil (Theil index and its decomposition),
- ratio: ratio (contribution rates of each component).

**Examples**

```
# Sample data
income = c(10, 20, 30, 40, 100)
pop = c(100, 150, 200, 250, 300)

# Gini coefficient (individual data)
gini0(income)

# Gini coefficient (grouped data)
gini(income, pop)

# Theil index (individual sample)
data = data.frame(g = c("A", "A", rep("B", 10), rep("A", 6)),
                  y = c(10, 10, rep(8, 4), rep(6, 6), rep(4, 4), 2, 2))
theil0(data$y)

# Theil index (grouped average)
data2 = data |> dplyr::count(g, y, name = "pop")
theil(data2$y, data2$pop)

# Theil index with grouping (sample data)
theil0_g(data, "g", "y")

# Theil index with grouping (average data)
theil_g(data2, "g", "y", "pop")

# Theil index with two-level cross-grouping
data3 = data.frame(
  industry = c("A", "A", "A", "A", "B", "B", "B", "B"),
  area = c("East", "East", "West", "West", "East", "East", "West", "West"),
  province = c("Shanghai", "Beijing", "Sichuan", "Yunnan",
               "Shanghai", "Beijing", "Sichuan", "Yunnan"),
  avg_wage = c(500, 400, 80, 60, 300, 250, 50, 40),
  emp_num = c(100, 80, 90, 70, 120, 100, 80, 60)
)
theil_g2_cross(data3, "industry", "area", "avg_wage", "emp_num")

# Theil index with two-level nested grouping
data4 = data.frame(
  province = c("A", "A", "A", "A", "B", "B"),
  city = c("A1", "A1", "A2", "A2", "B1", "B1"),
  industry = c("Manu", "Serv", "Manu", "Serv", "Manu", "Serv"),
  y = c(50000, 45000, 60000, 55000, 70000, 65000),
  pop = c(10000, 8000, 15000, 12000, 10000, 8000)
)
theil_g2_nest(data4, "province", "city", "y", "pop")
```

---

|                |                                              |
|----------------|----------------------------------------------|
| interp_hermite | <i>Piecewise Cubic Hermite Interpolation</i> |
|----------------|----------------------------------------------|

---

**Description**

Shape-preserving (monotone) piecewise cubic Hermite interpolation using the Fritsch-Carlson method via `splinefun(method = "monoH.FC")`.

**Usage**

```
interp_hermite(x, y, xout)
```

**Arguments**

|      |                                                                        |
|------|------------------------------------------------------------------------|
| x    | Numeric vector of observed x-values (no duplicates, length $\geq 2$ ). |
| y    | Numeric vector of observed y-values.                                   |
| xout | Numeric vector of x-values where interpolation is desired.             |

**Value**

A tibble with columns x and y. The attribute method is set to "hermite".

**Examples**

```
x = c(1, 2, 3, 4, 5)
y = c(1, 2, 4, 7, 11)
interp_hermite(x, y, xout = c(1.5, 2.5, 3.5))
```

---

|               |                             |
|---------------|-----------------------------|
| interp_linear | <i>Linear Interpolation</i> |
|---------------|-----------------------------|

---

**Description**

Performs linear (piecewise) interpolation at specified points.

**Usage**

```
interp_linear(x, y, xout)
```

**Arguments**

|      |                                                            |
|------|------------------------------------------------------------|
| x    | Numeric vector of observed x-values (no duplicates).       |
| y    | Numeric vector of observed y-values, same length as x.     |
| xout | Numeric vector of x-values where interpolation is desired. |

**Value**

A tibble with columns x and y, containing the interpolated values at xout. The attribute method is set to "linear".



**Examples**

```
x = c(1, 2, 3, 4, 5)
y = c(2, 4, 1, 5, 3)
interp_linear(x, y, xout = c(1.5, 2.5, 3.5))
```

interp\_poly

*Polynomial Interpolation***Description**

Fits a polynomial of specified degree via `lm()` and evaluates it at `xout`.

**Usage**

```
interp_poly(x, y, xout, degree = 3)
```

**Arguments**

|                     |                                                            |
|---------------------|------------------------------------------------------------|
| <code>x</code>      | Numeric vector of observed x-values (no duplicates).       |
| <code>y</code>      | Numeric vector of observed y-values.                       |
| <code>xout</code>   | Numeric vector of x-values where interpolation is desired. |
| <code>degree</code> | Integer, degree of the polynomial. Minimum 1.              |

**Value**

A tibble with columns `x` and `y`. The attribute method is set to "poly".

**Examples**

```
x = c(1, 2, 3, 4, 5)
y = c(2, 4, 1, 5, 3)
interp_poly(x, y, xout = c(1.5, 2.5, 3.5), degree = 3)
```

interp\_spline

*Cubic Spline Interpolation***Description**

Performs cubic spline interpolation. When `spar` is `NULL`, uses `stats::spline()` with the FMM method. Otherwise uses `stats::smooth.spline()` for smoothing.

**Usage**

```
interp_spline(x, y, xout, spar = NULL)
```

**Arguments**

|      |                                                                                                        |
|------|--------------------------------------------------------------------------------------------------------|
| x    | Numeric vector of observed x-values (no duplicates, length $\geq 4$ ).                                 |
| y    | Numeric vector of observed y-values.                                                                   |
| xout | Numeric vector of x-values where interpolation is desired.                                             |
| spar | Smoothing parameter passed to <code>smooth.spline()</code> . NULL for standard (non-smoothing) spline. |

**Value**

A tibble with columns x and y. The attribute method is set to "spline" or "smooth\_spline" depending on the variant used.

**Examples**

```
x = c(1, 2, 3, 4, 5)
y = c(2, 4, 1, 5, 3)
interp_spline(x, y, xout = c(1.5, 2.5, 3.5))
interp_spline(x, y, xout = c(1.5, 2.5, 3.5), spar = 0.8)
```

is\_ts\_df

*Check Whether an Object Is a ts\_df***Description**

Check Whether an Object Is a ts\_df

**Usage**

```
is_ts_df(x)
```

**Arguments**

|   |                  |
|---|------------------|
| x | Object to check. |
|---|------------------|

**Value**

TRUE or FALSE.

**Examples**

```
x = ts_df(data.frame(time = 1:3, value = c(10, 12, 15)))
is_ts_df(x)
```

---

|            |                                  |
|------------|----------------------------------|
| linear_sum | <i>Linear weighted synthesis</i> |
|------------|----------------------------------|

---

**Description**

Linear weighted synthesis

**Usage**

```
linear_sum(data, w)
```

**Arguments**

|      |                                                                       |
|------|-----------------------------------------------------------------------|
| data | Indicator matrix or data frame (rows = samples, columns = indicators) |
| w    | Weight vector (length must match ncol(data))                          |

**Value**

Vector of comprehensive scores

**Examples**

```
data = data.frame(
  GDP = c(0.85, 0.72, 0.91, NA),
  Employment = c(0.78, 0.85, 0.67, 0.73),
  Environment = c(0.65, 0.72, NA, 0.81)
)
w = c(0.5, 0.3, 0.2)
linear_sum(data, w)
```

---

|              |                                |
|--------------|--------------------------------|
| markov_chain | <i>Markov Chain Prediction</i> |
|--------------|--------------------------------|

---

**Description**

Constructs a transition probability matrix from a state sequence and performs multi-step Markov chain prediction.

**Usage**

```
markov_chain(S, s0, n_steps = 5)
```

**Arguments**

|         |                                                         |
|---------|---------------------------------------------------------|
| S       | State sequence, supplied as a vector or factor.         |
| s0      | Initial state, must be one of the state levels in S.    |
| n_steps | Number of prediction steps. Must be a positive integer. |

## Details

For states that never appear as a starting state, equal transition probabilities are assigned to keep each row sum equal to 1. The ultimate stationary distribution is computed using eigenvalue decomposition.

## Value

A list with components:

**trans\_mat** Transition probability matrix.

**pred\_probs** Matrix of state probabilities for each prediction step.

**pred\_states** Predicted states, taking the most likely state at each step.

**pi\_final** Ultimate stationary distribution.

## Examples

```
# Weather states: Rainy, Cloudy, Sunny
S = factor(c("Sunny", "Sunny", "Cloudy", "Rainy", "Sunny",
             "Cloudy", "Sunny", "Sunny", "Rainy", "Cloudy",
             "Sunny", "Cloudy", "Rainy", "Sunny", "Sunny",
             "Cloudy", "Sunny", "Rainy", "Cloudy", "Sunny"),
           levels = c("Rainy", "Cloudy", "Sunny"))
markov_chain(S, s0 = "Cloudy", n_steps = 3)
```

---

membership

*Membership Functions for Fuzzy Logic*

---

## Description

A collection of functions to compute membership values for various fuzzy sets, including triangular, trapezoidal, Gaussian, generalized bell, two-parameter Gaussian, sigmoid, difference of sigmoids, product of sigmoids, Z-shaped, PI-shaped, and S-shaped membership functions. Includes a function to visualize membership functions using ggplot2. These are designed for evaluation models in mathematical modeling, compatible with `fuzzy_eval` in the `mathmodels` package.

## Usage

`tri_mf(x, params)`

`trap_mf(x, params)`

`gauss_mf(x, params)`

`gbell_mf(x, params)`

`gauss2mf(x, params)`

`sigmoid_mf(x, params)`

`dsigmoid_mf(x, params)`

```

psigmoid_mf(x, params)

z_mf(x, params)

pi_mf(x, params)

s_mf(x, params)

plot_mf(mf, xlim = c(0, 10), main = NULL)

```

### Arguments

|        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| x      | Numeric vector, input values for which to compute membership.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| params | Numeric vector, parameters defining the membership function: <ul style="list-style-type: none"> <li>• For <code>tri_mf</code>: <math>c(a, b, c)</math>, where <math>a \leq b \leq c</math> (left base, peak, right base).</li> <li>• For <code>trap_mf</code>: <math>c(a, b, c, d)</math>, where <math>a \leq b \leq c \leq d</math> (left base, left top, right top, right base).</li> <li>• For <code>gauss_mf</code>: <math>c(\text{sigma}, c)</math>, where <math>\text{sigma} &gt; 0</math> (spread, center).</li> <li>• For <code>gbell_mf</code>: <math>c(a, b, c)</math>, where <math>a &gt; 0, b &gt; 0</math> (width, shape, center).</li> <li>• For <code>gauss2mf</code>: <math>c(s1, c1, s2, c2)</math>, where <math>s1 &gt; 0, s2 &gt; 0</math> (left spread, left center, right spread, right center).</li> <li>• For <code>sigmoid_mf</code>: <math>c(a, b)</math>, where <math>a &gt; 0</math> (slope, inflection point).</li> <li>• For <code>dsigmoid_mf</code>: <math>c(a1, c1, a2, c2)</math>, where <math>a1 &gt; 0, a2 &gt; 0</math> (slopes and inflection points for two sigmoids).</li> <li>• For <code>psigmoid_mf</code>: <math>c(a1, c1, a2, c2)</math>, where <math>a1 &gt; 0, a2 &gt; 0</math> (slopes and inflection points for two sigmoids).</li> <li>• For <code>z_mf</code>: <math>c(a, b)</math>, where <math>a &lt; b</math> (left base, right base).</li> <li>• For <code>pi_mf</code>: <math>c(a, b, c, d)</math>, where <math>a &lt; b &lt; c &lt; d</math> (left base, left shoulder, right shoulder, right base).</li> <li>• For <code>s_mf</code>: <math>c(a, b)</math>, where <math>a &lt; b</math> (left base, right base).</li> </ul> |
| mf     | Function, a membership function with fixed parameters (e.g., <code>function(x) tri_mf(x, c(2, 5, 8))</code> ).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| xlim   | Numeric vector of length 2, x-axis limits for plotting (default $c(0, 10)$ ).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| main   | Character, plot title (default NULL, no title).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

### Details

These functions support evaluation models in mathematical modeling:

- `tri_mf`: Triangular membership, linear rise from  $a$  to  $b$  (peak) and fall to  $c$ .
- `trap_mf`: Trapezoidal membership, linear rise from  $a$  to  $b$ , plateau from  $b$  to  $c$ , fall to  $d$ .
- `gauss_mf`: Gaussian membership, bell-shaped curve centered at  $c$  with spread  $\text{sigma}$ .
- `gbell_mf`: Generalized bell membership, bell-shaped curve with width  $a$ , shape  $b$ , and center  $c$ .
- `gauss2mf`: Two-parameter Gaussian membership, combining two Gaussians with spreads  $s1, s2$  and centers  $c1, c2$ .
- `sigmoid_mf`: Sigmoid membership, S-shaped curve with slope  $a$  and inflection point  $b$ .

- `dsigmoid_mf`: Difference of two sigmoids, combining slopes `a1`, `a2` and inflection points `c1`, `c2`.
- `psigmoid_mf`: Product of two sigmoids, combining slopes `a1`, `a2` and inflection points `c1`, `c2`.
- `z_mf`: Z-shaped membership, decreasing from 1 at `a` to 0 at `b`.
- `pi_mf`: PI-shaped membership, rising from `a` to `b`, plateau from `b` to `c`, falling to `d`.
- `s_mf`: S-shaped membership, increasing from 0 at `a` to 1 at `b`.
- `plot_mf`: Plots a membership function over `xlim` using `ggplot2`, suitable for tidyverse workflows.

Membership values can be used to construct fuzzy evaluation matrices for `fuzzy_eval`. Implemented in base R, except `plot_mf`, which requires `ggplot2`.

### Value

- For membership functions (`tri_mf`, `trap_mf`, `gauss_mf`, `gbell_mf`, `gauss2mf`, `sigmoid_mf`, `dsigmoid_mf`, `psigmoid_mf`, `z_mf`, `pi_mf`, `s_mf`): A numeric vector of membership values in  $[0, 1]$ , same length as `x`.
- For `plot_mf`: A `ggplot2` object, plotting the membership function.

### Examples

```
# Define input values
x = 0:10

# Triangular membership
tri_mf(x, params = c(3, 6, 8))

# Trapezoidal membership
trap_mf(x, params = c(1, 5, 7, 8))

# Gaussian membership
gauss_mf(x, params = c(2, 5))

# Generalized bell membership
gbell_mf(x, params = c(2, 4, 6))

# Two-parameter Gaussian membership
gauss2mf(x, params = c(1, 3, 3, 4))

# Sigmoid membership
sigmoid_mf(x, params = c(2, 4))

# Difference of sigmoids membership
dsigmoid_y = dsigmoid_mf(x, params = c(5, 2, 5, 7))

# Product of sigmoids membership
psigmoid_mf(x, params = c(2, 3, -5, 8))

# Z-shaped membership
z_mf(x, params = c(3, 7))

# PI-shaped membership
pi_mf(x, params = c(1, 4, 5, 10))
```

```

# S-shaped membership
s_mf(x, params = c(1, 8))

## Not run:
# Visualize membership functions
plot_mf(\(x) tri_mf(x, c(3, 6, 8)), main = "Triangular MF")
plot_mf(\(x) trap_mf(x, c(1, 5, 7, 8)), main = "Trapezoidal MF")
plot_mf(\(x) gauss_mf(x, c(2, 5)), main = "Gaussian MF")
plot_mf(\(x) gbell_mf(x, c(2, 4, 6)), main = "Generalized Bell MF")
plot_mf(\(x) gauss2mf(x, c(1, 3, 3, 4)), main = "Two-Parameter Gaussian MF")
plot_mf(\(x) sigmoid_mf(x, c(2, 4)), main = "Sigmoid MF")
plot_mf(\(x) dsigmoid_mf(x, c(5, 2, 5, 7)), main = "Difference of Sigmoids MF")
plot_mf(\(x) psigmoid_mf(x, c(2, 3, -5, 8)), main = "Product of Sigmoids MF")
plot_mf(\(x) z_mf(x, c(3, 7)), main = "Z-Shaped MF")
plot_mf(\(x) pi_mf(x, c(1, 4, 5, 10)), main = "PI-Shaped MF")
plot_mf(\(x) s_mf(x, c(1, 8)), main = "S-Shaped MF")

## End(Not run)

```

---

model\_logistic

---

Logistic Population Growth Model

---

## Description

Solves the logistic growth equation:

$$\frac{dN}{dt} = r \left( 1 - \frac{N}{K} \right) N$$

## Usage

```
model_logistic(init, params, times, ...)
```

## Arguments

|        |                                                 |
|--------|-------------------------------------------------|
| init   | Named numeric vector, e.g. c(N = 10).           |
| params | Named numeric vector, e.g. c(r = 0.5, K = 100). |
|        | <b>r</b> Intrinsic growth rate.                 |
|        | <b>K</b> Carrying capacity.                     |
| times  | Numeric vector of output times.                 |
| ...    | Additional arguments passed to ode_solver.      |

## Details

The analytical solution is  $N(t) = K / (1 + (K/N_0 - 1) e^{-rt})$ , which can be used to verify numerical accuracy.

## Value

A data.frame with columns time and N.

**Examples**

```
res = model_logistic(
  init   = c(N = 10),
  params = c(r = 0.5, K = 100),
  times  = seq(0, 20, by = 0.1)
)
head(res)
```

model\_lv

*Lotka-Volterra Predator-Prey Model***Description**

Classic two-species predator-prey system:

$$\frac{dx}{dt} = \alpha x - \beta xy$$

$$\frac{dy}{dt} = \delta xy - \mu y$$

**Usage**

```
model_lv(init, params, times, ...)
```

**Arguments**

|                     |                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>init</code>   | Named numeric vector, e.g. <code>c(x = 40, y = 9)</code> .                                                                                                                                                                                                                                                                                                                            |
| <code>params</code> | Named numeric vector with four parameters:<br><b>alpha</b> Prey intrinsic growth rate.<br><b>beta</b> Predation rate (prey killed per predator per unit time).<br><b>delta</b> Predator reproduction rate per prey consumed.<br><b>mu</b> Predator mortality rate. Named <code>mu</code> (not <code>gamma</code> ) to avoid confusion with the recovery rate used in epidemic models. |
| <code>times</code>  | Numeric vector of output times.                                                                                                                                                                                                                                                                                                                                                       |
| <code>...</code>    | Additional arguments passed to <code>ode_solver</code> .                                                                                                                                                                                                                                                                                                                              |

**Details**

- `x` Prey population.
- `y` Predator population.

**Value**

A `data.frame` with columns `time`, `x`, `y`.



**Examples**

```
model_lv(
  init = c(x = 40, y = 9),
  params = c(alpha = 0.4, beta = 0.04, delta = 0.02, mu = 0.5),
  times = seq(0, 100, by = 0.1)
)
```

model\_malthus

*Malthusian (Exponential) Growth Model***Description**

Solves the Malthusian growth equation:

$$\frac{dN}{dt} = r N$$

**Usage**

```
model_malthus(init, params, times, ...)
```

**Arguments**

|        |                                                 |
|--------|-------------------------------------------------|
| init   | Named numeric vector, e.g. c(N = 100).          |
| params | Named numeric vector, e.g. c(r = 0.3).          |
|        | <b>r</b> Intrinsic growth rate (per time unit). |
| times  | Numeric vector of output times.                 |
| ...    | Additional arguments passed to ode_solver.      |

**Details**

The analytical solution is  $N(t) = N_0 e^{rt}$ , which can be used to verify numerical accuracy.

**Value**

A data.frame with columns time and N.

**Examples**

```
res = model_malthus(
  init = c(N = 100),
  params = c(r = 0.3),
  times = seq(0, 10, by = 0.1)
)
# Compare with analytical solution
res$N_exact = 100 * exp(0.3 * res$time)
head(res)
```

model\_seir

*SEIR Epidemic Model***Description**

Four-compartment model that adds an exposed (latent) class.

**Usage**

```
model_seir(init, params, times, ...)
```

**Arguments**

**init** Named numeric vector. R defaults to 0 if omitted, e.g. `c(S = 980, E = 10, I = 10)`.

**params** Named numeric vector, e.g. `c(beta = 0.3, alpha = 0.2, gamma = 0.1)`.

**beta** Transmission rate.

**alpha** Rate of progression from exposed to infectious ( $1/\alpha$  = mean latent period).

**gamma** Recovery rate ( $1/\gamma$  = mean infectious period).

**N** Total population size. Optional — defaults to `sum(init)`.

**b** Birth rate (per capita per time). Optional.

**d** Death rate (per capita per time). Optional.

**times** Numeric vector of output times.

**...** Additional arguments passed to `ode_solver`.

**Details**

**Closed population** (default):

$$\begin{aligned}\frac{dS}{dt} &= -\beta \frac{SI}{N} \\ \frac{dE}{dt} &= \beta \frac{SI}{N} - \alpha E \\ \frac{dI}{dt} &= \alpha E - \gamma I \\ \frac{dR}{dt} &= \gamma I\end{aligned}$$

**Open population** (when `b / d` are provided):

$$\begin{aligned}\frac{dS}{dt} &= bN - \beta \frac{SI}{N} - dS \\ \frac{dE}{dt} &= \beta \frac{SI}{N} - \alpha E - dE \\ \frac{dI}{dt} &= \alpha E - \gamma I - dI \\ \frac{dR}{dt} &= \gamma I - dR\end{aligned}$$

- **S** Susceptible — not yet infected.
- **E** Exposed — infected but not yet infectious (mean latent period  $1/\alpha$ ).
- **I** Infectious — actively spreading the pathogen (mean infectious period  $1/\gamma$ ).
- **R** Recovered — permanently immune.

In the closed case population size  $N = S + E + I + R$  is conserved.

### Value

A `data.frame` with columns `time`, `S`, `E`, `I`, `R`.

### Examples

```
# Closed population
model_seir(
  init   = c(S = 980, E = 10, I = 10),
  params = c(beta = 0.3, alpha = 0.2, gamma = 0.1),
  times  = seq(0, 50, by = 0.1)
)

# Open population with births and deaths
model_seir(
  init   = c(S = 980, E = 10, I = 10),
  params = c(beta = 0.3, alpha = 0.2, gamma = 0.1, b = 0.01, d = 0.01),
  times  = seq(0, 200, by = 1)
)
```

---

model\_si

*SI Epidemic Model*

---

### Description

Two-compartment model with no recovery.

### Usage

```
model_si(init, params, times, ...)
```

### Arguments

|                     |                                                                                                                                                                                                                                                                                                                                   |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>init</code>   | Named numeric vector, e.g. <code>c(S = 990, I = 10)</code> .                                                                                                                                                                                                                                                                      |
| <code>params</code> | Named numeric vector, e.g. <code>c(beta = 0.002)</code> .<br><b>beta</b> Transmission rate (contacts per individual per time).<br><b>N</b> Total population size. Optional — defaults to <code>sum(init)</code> .<br><b>b</b> Birth rate (per capita per time). Optional.<br><b>d</b> Death rate (per capita per time). Optional. |
| <code>times</code>  | Numeric vector of output times.                                                                                                                                                                                                                                                                                                   |
| <code>...</code>    | Additional arguments passed to <code>ode_solver</code> .                                                                                                                                                                                                                                                                          |

**Details****Closed population** (default):

$$\frac{dS}{dt} = -\beta \frac{SI}{N}$$

$$\frac{dI}{dt} = \beta \frac{SI}{N}$$

**Open population** (when b / d are provided):

$$\frac{dS}{dt} = bN - \beta \frac{SI}{N} - dS$$

$$\frac{dI}{dt} = \beta \frac{SI}{N} - dI$$

In the closed case population size  $N = S + I$  is conserved.**Value**

A data.frame with columns time, S, I.

**Examples**

```
# Closed population
model_si(
  init   = c(S = 990, I = 10),
  params = c(beta = 0.002),
  times  = seq(0, 30, by = 0.1)
)

# Open population with births and deaths
model_si(
  init   = c(S = 990, I = 10),
  params = c(beta = 0.002, b = 0.01, d = 0.01),
  times  = seq(0, 200, by = 1)
)
```

model\_sir

*SIR Epidemic Model***Description**

Three-compartment model with permanent immunity after recovery.

**Usage**

```
model_sir(init, params, times, ...)
```

**Arguments**

|                     |                                                                                                                                                                                                                                                                                                                                                                                         |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>init</code>   | Named numeric vector. R defaults to 0 if omitted, e.g. <code>c(S = 990, I = 10)</code> .                                                                                                                                                                                                                                                                                                |
| <code>params</code> | Named numeric vector, e.g. <code>c(beta = 0.002, gamma = 0.1)</code> .<br><b>beta</b> Transmission rate.<br><b>gamma</b> Recovery rate ( $1/\text{gamma}$ = mean infectious period).<br><b>N</b> Total population size. Optional — defaults to <code>sum(init)</code> .<br><b>b</b> Birth rate (per capita per time). Optional.<br><b>d</b> Death rate (per capita per time). Optional. |
| <code>times</code>  | Numeric vector of output times.                                                                                                                                                                                                                                                                                                                                                         |
| <code>...</code>    | Additional arguments passed to <code>ode_solver</code> .                                                                                                                                                                                                                                                                                                                                |

**Details****Closed population** (default):

$$\begin{aligned}\frac{dS}{dt} &= -\beta \frac{SI}{N} \\ \frac{dI}{dt} &= \beta \frac{SI}{N} - \gamma I \\ \frac{dR}{dt} &= \gamma I\end{aligned}$$

**Open population** (when `b / d` are provided):

$$\begin{aligned}\frac{dS}{dt} &= bN - \beta \frac{SI}{N} - dS \\ \frac{dI}{dt} &= \beta \frac{SI}{N} - \gamma I - dI \\ \frac{dR}{dt} &= \gamma I - dR\end{aligned}$$

In the closed case population size  $N = S + I + R$  is conserved. The basic reproduction number is  $R_0 = \beta/\gamma$ .

**Value**

A `data.frame` with columns `time`, `S`, `I`, `R`.

**Examples**

```
# Closed population
model_sir(
  init = c(S = 990, I = 10),
  params = c(beta = 0.002, gamma = 0.1),
  times = seq(0, 50, by = 0.1)
)

# Open population with births and deaths
model_sir(
  init = c(S = 990, I = 10),
  params = c(beta = 0.002, gamma = 0.1, b = 0.01, d = 0.01),
  times = seq(0, 200, by = 1)
)
```

model\_sis

*SIS Epidemic Model***Description**

Two-compartment model with recovery but no lasting immunity.

**Usage**

```
model_sis(init, params, times, ...)
```

**Arguments**

**init** Named numeric vector, e.g. `c(S = 990, I = 10)`.

**params** Named numeric vector, e.g. `c(beta = 0.002, gamma = 0.1)`.

**beta** Transmission rate.

**gamma** Recovery rate ( $1/\text{gamma}$  = mean infectious period).

**N** Total population size. Optional — defaults to `sum(init)`.

**b** Birth rate (per capita per time). Optional.

**d** Death rate (per capita per time). Optional.

**times** Numeric vector of output times.

**...** Additional arguments passed to `ode_solver`.

**Details**

**Closed population** (default):

$$\frac{dS}{dt} = -\beta \frac{SI}{N} + \gamma I$$

$$\frac{dI}{dt} = \beta \frac{SI}{N} - \gamma I$$

**Open population** (when `b` / `d` are provided):

$$\frac{dS}{dt} = bN - \beta \frac{SI}{N} + \gamma I - dS$$

$$\frac{dI}{dt} = \beta \frac{SI}{N} - \gamma I - dI$$

In the closed case population size  $N = S + I$  is conserved.

**Value**

A `data.frame` with columns `time`, `S`, `I`.

## Examples

```
# Closed population
model_sis(
  init   = c(S = 990, I = 10),
  params = c(beta = 0.002, gamma = 0.1),
  times  = seq(0, 50, by = 0.1)
)

# Open population with births and deaths
model_sis(
  init   = c(S = 990, I = 10),
  params = c(beta = 0.3, gamma = 0.1, b = 0.01, d = 0.01),
  times  = seq(0, 200, by = 1)
)
```

ode\_solver

*General ODE Solver*

## Description

Solves a system of ordinary differential equations (ODEs) using string-formula equations. Equation strings are parsed once at call time and reused across all integration steps, so the solver is substantially faster than approaches that call `parse()` inside the derivative function.

## Usage

```
ode_solver(init, times, equations, params = NULL, method = "lsoda", ...)
```

## Arguments

|                        |                                                                                                                                |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| <code>init</code>      | A named numeric vector of initial values for all state variables.                                                              |
| <code>times</code>     | A numeric vector of output times.                                                                                              |
| <code>equations</code> | A named character vector; names are variable names, values are derivative expressions (e.g. <code>c(S = "-beta*S*I")</code> ). |
| <code>params</code>    | A named numeric vector or list of parameters referenced inside equations.                                                      |
| <code>method</code>    | Integration method passed to <code>deSolve::ode</code> . Default <code>"lsoda"</code> .                                        |
| <code>...</code>       | Additional arguments forwarded to <code>deSolve::ode</code> .                                                                  |

## Details

All equations are parsed with `parse()` exactly once before the integrator starts. A single pre-allocated environment is reused on every derivative evaluation, avoiding repeated memory allocation and garbage collection.

## Value

A `data.frame` with a time column followed by one column per state variable.

## Examples

```
# Built-in wrapper (one-liner)
sir = model_sir(
  init = c(S = 990, I = 10),
  params = c(beta = 0.002, gamma = 0.1),
  times = seq(0, 50, by = 0.1)
)
head(sir)

# Custom SEIR model with demography
seir_demo = ode_solver(
  init = c(S = 1000, E = 1, I = 0, R = 0),
  times = seq(0, 200, by = 1),
  equations = c(
    S = "mu*N - beta*S*I - mu*S",
    E = "beta*S*I - alpha*E - mu*E",
    I = "alpha*E - gamma*I - mu*I",
    R = "gamma*I - mu*R"
  ),
  params = c(beta = 0.3, alpha = 0.2, gamma = 0.1, mu = 0.01, N = 1000)
)
tail(seir_demo)
```

pca\_weight

*PCA-Based Weighting Method*

## Description

Computes indicator weights using Principal Component Analysis (PCA). The method extracts principal components and uses their variance contribution to derive objective weights for indicators. Optionally handles positive/negative directions of indicators, and supports pre-standardized data.

## Usage

```
pca_weight(X, index = NULL, nfs = NULL, varimax = TRUE, method = "abs")
```

## Arguments

|         |                                                                                                                                                                                                                                                                                                                                                                                            |
|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| X       | A numeric data frame or matrix where rows represent samples and columns represent indicators.                                                                                                                                                                                                                                                                                              |
| index   | A character vector indicating the direction of each indicator. Use "+" for positive indicators (higher is better), "-" for negative indicators (lower is better), and NA for already standardized indicators (no standardization will be applied).<br><br>If <code>index = NULL</code> (default), all indicators are treated as <code>NA</code> , meaning no standardization is performed. |
| nfs     | Number of principal components to use; by default, all are used.                                                                                                                                                                                                                                                                                                                           |
| varimax | Whether to perform Varimax rotation, default is TRUE.                                                                                                                                                                                                                                                                                                                                      |
| method  | Weighting Method. "abs" uses absolute loading values $ a_{ji} $ (default), "squared" uses $a_{ji}^2$ .                                                                                                                                                                                                                                                                                     |



**Value**

A list containing:

|        |                                                           |
|--------|-----------------------------------------------------------|
| w      | Numeric vector of normalized weights for each indicator.  |
| s      | Numeric vector of scores for each sample.                 |
| lambda | Eigenvalues of principal components (explained variance). |
| varP   | Proportion of variance explained by selected PCs.         |

**Examples**

```
# Example: Using PCA to compute indicator weights
ind = c("+", "+", "-", "-")
pca_weight(iris[1:10, 1:4], ind, nfs = 2)
```

---

|                   |                                      |
|-------------------|--------------------------------------|
| plot_compartments | <i>Plot compartment trajectories</i> |
|-------------------|--------------------------------------|

---

**Description**

Draws one line per compartment over time. Use compartments to select a subset of the state variables; the default is to plot all columns except time.

**Usage**

```
plot_compartments(df, compartments = NULL)
```

**Arguments**

|              |                                                                                                             |
|--------------|-------------------------------------------------------------------------------------------------------------|
| df           | A data frame returned by ode_solver() or any built-in model function.                                       |
| compartments | Character vector of compartment names to plot. When NULL (the default) every column except time is plotted. |

**Value**

A [ggplot](#) object.

**Examples**

```
sir = model_sir(
  init = c(S = 990, I = 10),
  params = c(beta = 0.002, gamma = 0.1),
  times = seq(0, 50, by = 0.1)
)
plot_compartments(sir)
plot_compartments(sir, compartments = c("S", "I"))
```

---

|                |                                       |
|----------------|---------------------------------------|
| plot_incidence | <i>Plot daily new infections (dI)</i> |
|----------------|---------------------------------------|

---

### Description

Computes the daily change in the infectious compartment  $\Delta I$  directly via `diff()` and plots it as a line chart. The peak is highlighted with a point marker.

### Usage

```
plot_incidence(df)
```

### Arguments

|    |                                                                                                                                                |
|----|------------------------------------------------------------------------------------------------------------------------------------------------|
| df | A data frame returned by <code>ode_solver()</code> or any built-in model function. Must contain columns <code>time</code> and <code>I</code> . |
|----|------------------------------------------------------------------------------------------------------------------------------------------------|

### Value

A [ggplot](#) object.

### Examples

```
sir = model_sir(
  init   = c(S = 990, I = 10),
  params = c(beta = 0.002, gamma = 0.1),
  times  = seq(0, 50, by = 0.1)
)
plot_incidence(sir)
```

---

|               |                          |
|---------------|--------------------------|
| plot_phase_si | <i>Phase plot S vs I</i> |
|---------------|--------------------------|

---

### Description

Draws a trajectory in the  $(S, I)$  plane. Useful for visualising the epidemic orbit and threshold behaviour.

### Usage

```
plot_phase_si(df)
```

### Arguments

|    |                                                                                     |
|----|-------------------------------------------------------------------------------------|
| df | A data frame with columns <code>time</code> , <code>S</code> , and <code>I</code> . |
|----|-------------------------------------------------------------------------------------|

### Value

A [ggplot](#) object.

**Examples**

```
sir = model_sir(  
  init = c(S = 990, I = 10),  
  params = c(beta = 0.002, gamma = 0.1),  
  times = seq(0, 50, by = 0.1)  
)  
plot_phase_si(sir)
```

---

|                  |                        |
|------------------|------------------------|
| plot_reg_predict | <i>Prediction Plot</i> |
|------------------|------------------------|

---

**Description**

Returns a fitted-vs-observed line plot, with optional confidence bands.

**Usage**

```
plot_reg_predict(model_result, type = c("fitted", "predicted"), show_ci = TRUE)
```

**Arguments**

|              |                                                                 |
|--------------|-----------------------------------------------------------------|
| model_result | A fitted result from any reg_*() function.                      |
| type         | Character. Currently supports "fitted" only.                    |
| show_ci      | Logical. Whether to overlay confidence intervals. Default TRUE. |

**Value**

A [ggplot](#) object.

---

|                    |                             |
|--------------------|-----------------------------|
| plot_reg_residuals | <i>Residual Diagnostics</i> |
|--------------------|-----------------------------|

---

**Description**

Returns a multi-panel diagnostic plot from a fitted regression model.

**Usage**

```
plot_reg_residuals(model_result)
```

**Arguments**

|              |                                          |
|--------------|------------------------------------------|
| model_result | A list returned by any reg_*() function. |
|--------------|------------------------------------------|

**Value**

A ggplot-based composite.

---

|                  |                                                            |
|------------------|------------------------------------------------------------|
| plot_Rt_estimate | <i>Plot effective reproduction number <math>R_t</math></i> |
|------------------|------------------------------------------------------------|

---

### Description

Estimates the time-varying effective reproduction number  $R_t$  directly from compartment data. Two methods are available:

"mechanistic" Uses  $R_t = \beta S(t)/(\gamma N)$ . This is the standard formula for a frequency-dependent SIR-type model.

"normalized" Uses  $R_t = R_0 S(t)/N$  with  $R_0 = \beta/\gamma$ . Suitable when different normalisations are desired.

### Usage

```
plot_Rt_estimate(df, params, N = NULL, method = c("mechanistic", "normalized"))
```

### Arguments

|        |                                                                                                                         |
|--------|-------------------------------------------------------------------------------------------------------------------------|
| df     | A data frame with columns time and S.                                                                                   |
| params | A named numeric vector or list containing at least beta and gamma.                                                      |
| N      | Total population size. If NULL (the default), it is estimated from the sum of all state columns at the first time step. |
| method | Estimation method: "mechanistic" (default) or "normalized".                                                             |

### Value

A [ggplot](#) object with a dashed horizontal line at  $R_t = 1$ .

### Examples

```
sir = model_sir(
  init   = c(S = 990, I = 10),
  params = c(beta = 0.002, gamma = 0.1),
  times  = seq(0, 50, by = 0.1)
)
plot_Rt_estimate(sir, params = c(beta = 0.002, gamma = 0.1))
```

---

|         |                           |
|---------|---------------------------|
| plot_ts | <i>Plot a Time Series</i> |
|---------|---------------------------|

---

### Description

Plot a Time Series

### Usage

```
plot_ts(x, title = "Time Series", x_lab = "Time", y_lab = "Value")
```

**Arguments**

|              |              |
|--------------|--------------|
| x            | A ts_df.     |
| title        | Plot title.  |
| x_lab, y_lab | Axis labels. |

**Value**

A ggplot object.

**Examples**

```
plot_ts(as_ts_df(AirPassengers))
```

---

|             |                             |
|-------------|-----------------------------|
| plot_ts_acf | <i>Autocorrelation Plot</i> |
|-------------|-----------------------------|

---

**Description**

Autocorrelation Plot

**Usage**

```
plot_ts_acf(x, max_lag = 40L, title = "ACF")
```

**Arguments**

|         |                                          |
|---------|------------------------------------------|
| x       | A complete ts_df with no missing values. |
| max_lag | Maximum lag.                             |
| title   | Plot title.                              |

**Value**

A ggplot object.

**Examples**

```
plot_ts_acf(as_ts_df(log(AirPassengers)))
```

---

|                  |                       |
|------------------|-----------------------|
| plot_ts_forecast | <i>Plot Forecasts</i> |
|------------------|-----------------------|

---

**Description**

Plot Forecasts

**Usage**

```
plot_ts_forecast(
  x,
  forecast_tbl,
  level = c(80, 95),
  by = NULL,
  title = "Forecast"
)
```

**Arguments**

|              |                                         |
|--------------|-----------------------------------------|
| x            | Original ts_df.                         |
| forecast_tbl | Result from ts_forecast().              |
| level        | Confidence levels to draw when present. |
| by           | Step size for future time values.       |
| title        | Plot title.                             |

**Value**

A ggplot object.

**Examples**

```
fit = ts_ets(as_ts_df(log(AirPassengers)))
fc = ts_forecast(fit, h = 3)
plot_ts_forecast(as_ts_df(log(AirPassengers)), fc, by = 1)
```

---

|              |                                     |
|--------------|-------------------------------------|
| plot_ts_pacf | <i>Partial Autocorrelation Plot</i> |
|--------------|-------------------------------------|

---

**Description**

Partial Autocorrelation Plot

**Usage**

```
plot_ts_pacf(x, max_lag = 40L, title = "PACF")
```

**Arguments**

|         |                                          |
|---------|------------------------------------------|
| x       | A complete ts_df with no missing values. |
| max_lag | Maximum lag.                             |
| title   | Plot title.                              |

**Value**

A ggplot object.

**Examples**

```
plot_ts_pacf(as_ts_df(log(AirPassengers)))
```

---

|                   |                                 |
|-------------------|---------------------------------|
| plot_ts_residuals | <i>Residual Diagnostic Plot</i> |
|-------------------|---------------------------------|

---

**Description**

Residual Diagnostic Plot

**Usage**

```
plot_ts_residuals(model_result, max_lag = 30L, title = "Residual Diagnostics")
```

**Arguments**

|              |                                      |
|--------------|--------------------------------------|
| model_result | Result from ts_ets() or ts_sarima(). |
| max_lag      | Maximum lag.                         |
| title        | Plot title.                          |

**Value**

A patchwork plot.

**Examples**

```
fit = ts_sarima(as_ts_df(log(AirPassengers)), stepwise = TRUE, approximation = TRUE)
plot_ts_residuals(fit)
```

---

|             |                               |
|-------------|-------------------------------|
| plot_ts_stl | <i>STL Decomposition Plot</i> |
|-------------|-------------------------------|

---

**Description**

STL Decomposition Plot

**Usage**

```
plot_ts_stl(stl_result, title = "STL Decomposition")
```

**Arguments**

|            |                       |
|------------|-----------------------|
| stl_result | Result from ts_stl(). |
| title      | Plot title.           |

**Value**

A patchwork plot.

**Examples**

```
plot_ts_stl(ts_stl(as_ts_df(AirPassengers)))
```

---

|          |                       |
|----------|-----------------------|
| poly_fit | <i>Polynomial Fit</i> |
|----------|-----------------------|

---

**Description**

Fits a polynomial regression model via lm().

**Usage**

```
poly_fit(x, y, degree = 1)
```

**Arguments**

|        |                                                           |
|--------|-----------------------------------------------------------|
| x      | Numeric vector of observed predictor values.              |
| y      | Numeric vector of observed response values.               |
| degree | Integer, degree of the polynomial. Default is 1 (linear). |

**Value**

A named list with elements: model (lm object), coefficient (tibble with estimates, SE, t-value, p-value, 95% CI), model\_info (tibble with r.squared, adj.r.squared, AIC, BIC, sigma), residuals (tibble of .observed, .fitted, .residual), formula (character), type (character), and input (tibble of x, y).



**Examples**

```
x = 1:20
y = 3 + 2*x + rnorm(20, sd = 2)
poly_fit(x, y, degree = 1)
poly_fit(x, y, degree = 2)
```

preprocess

*Preprocessing Functions for Data Normalization and Standardization***Description**

A collection of functions to preprocess numeric data, including standardization, L2 norm normalization, Min-Max scaling, centered-type normalization, interval-type normalization, extreme-value-based normalization, initial-value-based normalization, mean-based normalization, and negative-to-positive transformation. These functions transform a numeric vector to a standardized or normalized scale, suitable for various indicator types (positive, negative, centered, interval-based, or extreme-based).

**Usage**

```
standardize(x, center = TRUE, scale = TRUE)

normalize(x)

rescale(x, type = "+", a = 0, b = 1)

rescale_middle(x, m)

rescale_interval(x, a, b)

rescale_extreme(x, type = "+")

rescale_initial(x, type = "+")

rescale_mean(x)

to_positive(x, type = "minmax")
```

**Arguments**

|                     |                                                                                                                                                                                                                                   |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>x</code>      | Numeric vector to be preprocessed.                                                                                                                                                                                                |
| <code>center</code> | Logical or numeric scalar, passed to <code>base::scale</code> for centering (for <code>standardize</code> ). Default is <code>TRUE</code> .                                                                                       |
| <code>scale</code>  | Logical or numeric scalar, passed to <code>base::scale</code> for scaling (for <code>standardize</code> ). Default is <code>TRUE</code> .                                                                                         |
| <code>type</code>   | Character scalar specifying the transformation direction or method:<br><code>"+"</code> Positive direction (larger values are better, for <code>rescale</code> , <code>rescale_extreme</code> and <code>rescale_initial</code> ). |

|   |                                                                                                                                    |
|---|------------------------------------------------------------------------------------------------------------------------------------|
|   | "-" Negative direction (smaller values are better, for <code>rescale_extreme</code> and <code>rescale_initial</code> ).            |
|   | " <b>minmax</b> " Min-max transformation (for <code>to_positive</code> ).                                                          |
|   | " <b>reciprocal</b> " Reciprocal transformation (for <code>to_positive</code> ).                                                   |
| a | Numeric scalar, lower bound of the output range or optimal interval (for <code>rescale</code> and <code>rescale_interval</code> ). |
| b | Numeric scalar, upper bound of the output range or optimal interval (for <code>rescale</code> and <code>rescale_interval</code> ). |
| m | Numeric scalar, optimal value for centered-type normalization (for <code>rescale_middle</code> ).                                  |

## Details

These functions support various preprocessing needs in data analysis:

- `standardize`: Applies Z-score standardization (mean = 0, sd = 1), ideal for equalizing variances or normally distributed data.
- `normalize`: Scales the vector to unit length by dividing by its L2 (Euclidean) norm, useful for machine learning or similarity calculations.
- `rescale`: Performs Min-Max scaling to a specified range (default  $[0, 1]$ ), supporting positive or negative indicators.
- `rescale_middle`: Normalizes centered-type indicators, where values closer to an optimal value `m` are better, mapping to  $[0, 1]$ .
- `rescale_interval`: Normalizes interval-type indicators, where values within  $[a, b]$  are optimal, mapping to  $[0, 1]$ .
- `rescale_extreme`: Normalizes using extreme values:  $\min(x)/x$  for positive indicators or  $x/\max(x)$  for negative indicators, often used in grey relational analysis.
- `rescale_initial`: Normalizes by dividing by the first value ( $x/x[1]$  or  $x[1]/x$ ), commonly used in grey relational analysis.
- `rescale_mean`: Normalizes by dividing by the mean ( $x/\text{mean}(x)$ ), commonly used in grey relational analysis.
- `to_positive`: Converts negative indicators to positive using either min-max ( $\max(x) - x$ ) or reciprocal ( $1/x$ ) transformation.

Missing values (NA) are preserved in the output. For `rescale_initial` and `rescale_mean`, the initial value or mean must be non-zero, respectively.

## Value

A numeric vector of the same length as `x`, transformed as follows:

- `standardize`: Standardized values (mean = 0, sd = 1).
- `normalize`: L2 norm normalized values (Euclidean norm, unit length).
- `rescale`: Min-Max scaled values in  $[a, b]$  (default  $[0, 1]$ ).
- `rescale_middle`: Centered-type normalized values in  $[0, 1]$ , where 1 indicates  $x = m$ .
- `rescale_interval`: Interval-type normalized values in  $[0, 1]$ , where 1 indicates  $x$  in  $[a, b]$ .
- `rescale_extreme`: Extreme-based normalized values using  $\min(x)/x$  (positive) or  $x/\max(x)$  (negative).
- `rescale_initial`: Initial-based normalized values using  $x/x[1]$  or  $x[1]/x$ .
- `rescale_mean`: Mean-based normalized values using  $x/\text{mean}(x)$ .
- `to_positive`: Transformed values converting negative indicators to positive using min-max or reciprocal transformation.

**Examples**

```

# Standardization
x = c(4, 1, NA, 5, 8)
standardize(x)

# L2 norm normalization
normalize(x)

# Min-Max normalization (positive direction)
rescale(x)          # Scale to \code{[0, 1]}
rescale(x, type = "-", a = 0.002, b = 0.996) # Reverse scaling

# Negative-to-positive transformation
to_positive(x)      # Min-max transformation
to_positive(x, type = "reciprocal") # Reciprocal transformation

# Centered-type normalization
PH = 6:9
rescale_middle(PH, 7)

# Interval-type normalization
Temp = c(35.2, 35.8, 36.6, 37.1, 37.8, 38.4)
rescale_interval(Temp, 36, 37)

# Extreme-based normalization
rescale_extreme(x)    # min(x)/x
rescale_extreme(x, "-") # x/max(x)

# Initial-based normalization
rescale_initial(x)

# Mean-based normalization
rescale_mean(x)

```

rank\_sum\_ratio

*Rank Sum Ratio (RSR) Evaluation***Description**

Performs Rank Sum Ratio (RSR) evaluation on a dataset of positive indicators, computing ranks, weighted RSR values, and a linear regression model to fit RSR against probit-transformed ranks. Supports integer or non-integer ranking methods.

**Usage**

```
rank_sum_ratio(data, w = NULL, method = "int")
```

**Arguments**

|      |                                                                                                           |
|------|-----------------------------------------------------------------------------------------------------------|
| data | Data frame with positive indicator data; first column is an ID column for identifying evaluation objects. |
| w    | Numeric vector, weights for indicators (default = equal weights).                                         |

method                      Character scalar, ranking method: "int" for integer ranks or "non-int" for scaled ranks in [1, n] (default = "int").

### Details

The `rank_sum_ratio` function implements the RSR method for evaluating objects based on positive indicators. It ranks the indicators (using integer or non-integer methods), computes weighted RSR values, adjusts ranks with probit transformation, and fits a linear regression model to relate RSR to probit values. The function assumes the first column of data is an ID column, and weights (*w*) can be provided or set to equal weights by default.

### Value

A list containing:

- `resultTable`: Data frame with RSR values, ranks, cumulative frequencies, probit values, and fitted RSR values.
- `reg`: Linear model object fitting RSR against probit values.
- `rankTable`: Data frame with ranked indicator values.

### Examples

```
# Example data
data = data.frame(ID = c("A", "B", "C"), X1 = c(10, 20, 15), X2 = c(5, 10, 8))
w = c(0.4, 0.6)
rank_sum_ratio(data, w, method = "int")
```

---

read\_nbs

*Read and Combine National Bureau of Statistics XLS Files*

---

### Description

This function reads multiple XLS files downloaded from the National Bureau of Statistics of China website (<https://www.stats.gov.cn>) without requiring renaming. It extracts the variable name from the "indicator: XXX" text in cell A1 of each file, skips the first 3 rows of headers, reads the first 31 rows of data, and combines the data into a single data frame by stacking vertically based on the indicator names. It also simplifies region names by removing suffixes such as "city", "province", "autonomous region", "clan", or "Uyghur".

### Usage

```
read_nbs(paths)
```

### Arguments

paths                      A character vector containing the file paths to the XLS files.

### Value

A tibble containing the combined data with an additional column `indicator` representing the original indicator names and a `region` column with simplified region names.

## Examples

```
## Not run:
paths = c("file1.xls", "file2.xls")
data = read_nbs(paths)

## End(Not run)
```

---

|                    |                                     |
|--------------------|-------------------------------------|
| regional_economics | <i>Regional Economics Functions</i> |
|--------------------|-------------------------------------|

---

## Description

A collection of functions for calculating regional economics indices, including Location Quotient (LQ), Herfindahl-Hirschman Index (HHI), and Ellison-Glaeser Index (EG). These functions are designed for analyzing regional and industrial data to assess spatial concentration and market structure.

## Usage

```
LQ(data, region, total, .cols, .by = NULL)
```

```
HHI(x, scaled = FALSE)
```

```
EG(data, region, industry, y)
```

## Arguments

|          |                                                                                                                                                                                                                                                                                                                       |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| data     | A data frame containing the necessary data for calculations. For LQ, the first row is assumed to be the national total, with the first two columns specifying the region and total, optionally including a grouping column. For EG, it contains region, industry, and indicator columns (e.g., employment or output). |
| region   | The name of the column in data specifying the region (used in LQ and EG).                                                                                                                                                                                                                                             |
| total    | The name of the column in data specifying the total (used in LQ).                                                                                                                                                                                                                                                     |
| .cols    | The columns in data for which to calculate the location quotient (used in LQ).                                                                                                                                                                                                                                        |
| .by      | Optional grouping column for LQ, defaults to NULL (no grouping).                                                                                                                                                                                                                                                      |
| x        | A numeric vector for calculating the HHI (used in HHI).                                                                                                                                                                                                                                                               |
| scaled   | Logical; if TRUE, the HHI is scaled to account for the number of firms. Defaults to FALSE.                                                                                                                                                                                                                            |
| industry | The name of the column in data specifying the industry (used in EG).                                                                                                                                                                                                                                                  |
| y        | The name of the column in data specifying the indicator (e.g., employment or output, used in EG).                                                                                                                                                                                                                     |

## Details

**LQ** Calculates the Location Quotient for multiple columns with optional grouping. The LQ measures the relative concentration of an industry in a region compared to a national benchmark. The function assumes the first row of data contains national totals, with the first two columns specifying the region and total, and the remaining columns used for LQ calculation.

**HHI** Calculates the Herfindahl-Hirschman Index, a measure of market concentration based on the squared sum of market shares. If `scaled = TRUE`, the HHI is normalized to account for the number of firms.

**EG** Calculates the Ellison-Glaeser Index, which measures the geographic concentration of an industry while controlling for firm size distribution and random distribution effects. It requires data on regions, industries, and an indicator (e.g., employment or output).

### Value

**LQ** A data frame with the region column and calculated location quotients for the specified columns, optionally grouped by `.by`.

**HHI** A numeric value representing the HHI, either scaled or unscaled.

**EG** A tibble with two columns: the industry name and the corresponding EG index value.

### Examples

```
# Example data
data = data.frame(
  region = c("National", "Region_A", "Region_B"),
  total = c(10000, 4000, 6000),
  industry1 = c(2000, 1000, 1000),
  industry2 = c(6000, 2000, 4000)
)

# Calculate Location Quotient
LQ(data, region, total, starts_with("industry"))

# Calculate HHI
x = c(50, 30, 20)
# Calculate the raw HHI
HHI(x)
# Calculate the standard (scaled) HHI
HHI(x, scaled = TRUE)

# Example data for EG
eg_data = data.frame(
  region = c("R1", "R1", "R1", "R1", "R2", "R2", "R3", "R3", "R1", "R2", "R2", "R3"),
  industry = c("A", "A", "A", "A", "A", "A", "A", "A", "B", "B", "B", "B"),
  employment = c(250, 200, 150, 100, 20, 15, 10, 5, 50, 200, 150, 50)
)
EG(eg_data, region, industry, y = employment)
```

---

reg\_diagnostics

*Model Diagnostics*


---

### Description

Runs diagnostic tests on a fitted regression model.

### Usage

```
reg_diagnostics(model_result)
```

**Arguments**

`model_result` A list returned by any `reg_*`() function.

**Value**

A named list with diagnostic tibbles. For linear models: VIF, Shapiro-Wilk normality, Breusch-Pagan, Durbin-Watson, residual stats. For logistic/poisson/negative binomial: Hosmer-Lemeshow / dispersion / deviance stats.

---

|        |                                        |
|--------|----------------------------------------|
| reg_lm | <i>Multivariable Linear Regression</i> |
|--------|----------------------------------------|

---

**Description**

Fits an OLS model via `lm()`, optionally performing stepwise selection using Akaike's information criterion.

**Usage**

```
reg_lm(
  formula,
  data,
  step = FALSE,
  direction = c("both", "forward", "backward"),
  ...
)
```

**Arguments**

`formula` A formula, e.g. `y ~ x1 + x2`.  
`data` A data frame containing the variables in formula.  
`step` Logical. Whether to perform stepwise selection (TRUE/FALSE). Default is FALSE.  
`direction` Character: "both" (default), "forward", or "backward".  
`...` Additional arguments passed to `lm()` and `step()`.

**Details**

When `step = TRUE`, the function first fits a null model (response only), then uses `step()` to search for the best combination of terms by AIC.

**Value**

A named list:

**model** Fitted `lm` object.

**coefficient** Tibble of coefficients, standard errors, t-values, p-values, 95% CIs.

**model\_info** Tibble of fit statistics (`r.squared`, `adj.r.squared`, AIC, BIC).

**residuals** Tibble of observed, fitted, residuals.

**diagnostics** Tibble of key fit metrics (Breusch-Pagan, Durbin-Watson).

**formula** Original formula.

**input** The original data frame.

## Examples

```
set.seed(42)
x1 = rnorm(100); x2 = rnorm(100)
y = 1 + 2*x1 - 3*x2 + rnorm(100, sd = 2)
reg_lm(y ~ x1 + x2, data = data.frame(y, x1, x2))
```

---

reg\_logistic

*Logistic Regression*


---

## Description

Fits a binary logistic regression model via `glm(family = binomial())`, with optional stepwise selection.

## Usage

```
reg_logistic(
  formula,
  data,
  step = FALSE,
  direction = c("both", "forward", "backward"),
  ...
)
```

## Arguments

|           |                                                                                |
|-----------|--------------------------------------------------------------------------------|
| formula   | A formula, e.g. $y \sim x1 + x2$ .                                             |
| data      | A data frame containing the variables in formula.                              |
| step      | Logical. Whether to perform stepwise selection (TRUE/FALSE). Default is FALSE. |
| direction | Character: "both" (default), "forward", or "backward".                         |
| ...       | Additional arguments passed to <code>lm()</code> and <code>step()</code> .     |

## Details

Same stepwise logic as [reg\\_lm](#), but uses GLM with the binomial family. A Hosmer-Lemeshow goodness-of-fit test is included in diagnostics.

## Value

A named list analogous to `reg_lm()`, plus:

**odds\_ratio** Tibble with odds ratios and 95% CIs.

**residuals** DataFrame of observed, fitted probability, residual.



**Examples**

```
set.seed(42)
x1 = rnorm(100)
logit = 0.5 + x1
prob = 1 / (1 + exp(-logit))
y = rbinom(100, 1, prob)
reg_logistic(y ~ x1, data = data.frame(y, x1))
```

---

|            |                                     |
|------------|-------------------------------------|
| reg_negbin | <i>Negative Binomial Regression</i> |
|------------|-------------------------------------|

---

**Description**

Fits a negative binomial regression model (for over-dispersed counts) via MASS::glm.nb().

**Usage**

```
reg_negbin(formula, data, step = FALSE, ...)
```

**Arguments**

|         |                                                |
|---------|------------------------------------------------|
| formula | A formula, e.g. count ~ x1 + x2.               |
| data    | A data frame.                                  |
| step    | Logical. Always FALSE for negative binomial.   |
| ...     | Additional arguments passed to MASS::glm.nb(). |

**Details**

**Note:** The glm.nb family does not support stepwise selection. Set step = FALSE.

For comparison with Poisson, inspect the dispersion ratio. NB is appropriate when dispersion » 1.

**Value**

A named list analogous to reg\_poisson(), plus:

**theta** Estimated dispersion parameter for the negative binomial distribution.

**residuals** DataFrame of observed, fitted, Pearson residual.

**Examples**

```
set.seed(42)
x1 = rnorm(100)
mu = exp(1 + 0.3*x1)
y = rnbinom(100, size = 1, mu = mu)
reg_negbin(y ~ x1, data = data.frame(y, x1))
```

---

`reg_poisson`*Poisson Regression*

---

### Description

Fits a count-response regression model via `glm(family = poisson())`, with optional stepwise selection.

### Usage

```
reg_poisson(  
  formula,  
  data,  
  step = FALSE,  
  direction = c("both", "forward", "backward"),  
  ...  
)
```

### Arguments

|                        |                                                                                |
|------------------------|--------------------------------------------------------------------------------|
| <code>formula</code>   | A formula, e.g. $y \sim x1 + x2$ .                                             |
| <code>data</code>      | A data frame containing the variables in formula.                              |
| <code>step</code>      | Logical. Whether to perform stepwise selection (TRUE/FALSE). Default is FALSE. |
| <code>direction</code> | Character: "both" (default), "forward", or "backward".                         |
| <code>...</code>       | Additional arguments passed to <code>lm()</code> and <code>step()</code> .     |

### Details

Performs an overdispersion check:  $\text{dispersion} > 1.5$  suggests NB may be preferable.

### Value

A named list analogous to `reg_lm()`, plus:

**dispersion** Dispersion ratio (should be near 1).

**residuals** Data frame with observed, fitted, Pearson residual.

### Examples

```
set.seed(42)  
x1 = rnorm(100)  
mu = exp(1 + 0.3*x1)  
y = rpois(100, mu)  
reg_poisson(y ~ x1, data = data.frame(y, x1))
```

---

|             |                    |
|-------------|--------------------|
| reg_predict | <i>Predictions</i> |
|-------------|--------------------|

---

### Description

Generates point predictions and confidence/ prediction intervals for new data.

### Usage

```
reg_predict(model_result, n_new = 10, level = 0.95)
```

### Arguments

|                           |                                                                                                                           |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------|
| <code>model_result</code> | A fitted result from any <code>reg_*</code> () function.                                                                  |
| <code>n_new</code>        | Number of new observations to predict. Random values drawn from the fitted model's residuals are used for predictor data. |
| <code>level</code>        | Confidence/interval level. Defaults to 0.95.                                                                              |

### Value

A named list with:

**predictions** Tibble with predicted values.

**confidence\_interval** Tibble with lower and upper bounds.

---

|                   |                                                                       |
|-------------------|-----------------------------------------------------------------------|
| system_evaluation | <i>System Evaluation Functions for Coupling and Obstacle Analysis</i> |
|-------------------|-----------------------------------------------------------------------|

---

### Description

These functions provide tools for system-level evaluation in multi-indicator systems:

- `coupling_degree()`: Computes coupling degree, coordination index, and coupling coordination degree for subsystems.
- `obstacle_degree()`: Computes obstacle degrees for secondary indicators to identify key constraints in the system, enabling batch processing with tidyverse for grouping and summarization.

### Usage

```
coupling_degree(data, w = NULL, id_cols = NULL, type = "standard")
```

```
obstacle_degree(data, w = NULL, id_cols = NULL, scaled = FALSE)
```

## Arguments

|                      |                                                                                                                                                                                                                                                                                                                                                                              |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>data</code>    | A data frame with normalized scores (usually in $[0, 1]$ ) as columns.                                                                                                                                                                                                                                                                                                       |
| <code>w</code>       | Optional vector of weights for indicators or subsystems; defaults to equal weights if NULL.                                                                                                                                                                                                                                                                                  |
| <code>id_cols</code> | Optional character vector of column names in <code>data</code> to preserve as identifiers (not used in calculations).                                                                                                                                                                                                                                                        |
| <code>type</code>    | Either "standard" for the standard coupling formula (results concentrated near 1) or "adjusted" for the revised formula (results more uniformly distributed in $[0, 1]$ ), as proposed by Wang Shujia, Kong Wei, et al. in "Misconceptions and Corrections of Domestic Coupling Coordination Degree Models, Journal of Natural Resources, 2021, 36(3): 793–810 (In Chinese)" |
| <code>scaled</code>  | Logical. Whether to perform row normalization on obstacle degrees (default FALSE).                                                                                                                                                                                                                                                                                           |

## Value

A tibble depending on the function:

**coupling\_degree** A tibble with columns:

- ID: Identifier columns specified by `id_cols` (if provided).
- coupling: Coupling Degree (range 0-1).
- coord: Coordination Index (range 0-1).
- coupling\_coord: Coupling Coordination Degree (range 0-1).

**obstacle\_degree** A tibble with:

- ID: Identifier columns specified by `id_cols` (if provided).
- Columns for secondary indicator obstacle degrees ( $O_{ij} = (1 - X_{ij}) * w_{ij}$ ).

Suitable for grouping and summarizing (e.g., with tidyverse) to compute primary indicator obstacle degrees ( $(U_i)$ ).

## Examples

```
# Sample normalized subsystem scores
df = data.frame(
  ID = LETTERS[1:6],
  s1 = c(0.0162, 0.1782, 0.5490, 0.6730, 0.0207, 0.9875),
  s2 = c(0.2720, 0.6824, 0.0593, 0.4812, 0.8891, 0.5573),
  s3 = c(0.2655, 0.3721, 0.5729, 0.9082, 0.2017, 0.8984)
)
# Coupling Degree Analysis
coupling_degree(df, id_cols = "ID")          # Equal weights
coupling_degree(df, c(0.4, 0.3, 0.3), id_cols = "ID",
  type = "adjusted")          # "adjusted" coupling degree
# Obstacle Degree Analysis
obstacle_degree(df, id_cols = "ID")          # Equal weights
obstacle_degree(df, c(0.4, 0.3, 0.3), id_cols = "ID")
```

topsis

*TOPSIS Method for Multi-Criteria Decision Making***Description**

Implements the Technique for Order of Preference by Similarity to Ideal Solution (TOPSIS) to rank alternatives based on multiple criteria. The function normalizes the decision matrix using L2 norm, applies weights, and computes relative closeness to the ideal solution.

**Usage**

```
topsis(X, w = NULL, index = NULL)
```

**Arguments**

|                    |                                                                                                                                                                                                                                              |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>X</code>     | A numeric matrix or data frame where rows represent alternatives and columns represent criteria.                                                                                                                                             |
| <code>w</code>     | A numeric vector of weights for each criterion. Must be non-negative and sum to 1. If not provided, equal weights are used.                                                                                                                  |
| <code>index</code> | A character vector indicating the direction of each indicator: Use "+" for positive indicators (higher is better), "-" for negative indicators (lower is better). If <code>index = NULL</code> (default), all indicators are treated as "+". |

**Details**

The TOPSIS method ranks alternatives by:

1. Normalizing the decision matrix using L2 norm normalization.
2. Applying weights to form a weighted normalized matrix.
3. Identifying positive and negative ideal solutions based on indicator directions.
4. Computing Euclidean distances to ideal solutions.
5. Calculating relative closeness as  $S_0 / (S_0 + S_{star})$ , where  $S_0$  is the distance to the negative ideal and  $S_{star}$  is the distance to the positive ideal.

This implementation supports both positive and negative indicators via the `index` parameter.

**Value**

A named numeric vector of relative closeness scores (in  $[0, 1]$ ) for each alternative. Higher values indicate better alternatives. Names are taken from `rownames(X)` or default to "Sample1", "Sample2", etc.

**Examples**

```
A = data.frame(
  X1 = c(2, 5, 3), # "+"
  X2 = c(8, 1, 6) # "-"
)
w = c(0.6, 0.4)
idx = c("+", "-")
topsis(A, w, idx)
```

---

|           |                             |
|-----------|-----------------------------|
| ts_arimax | <i>ARIMAX Model Fitting</i> |
|-----------|-----------------------------|

---

**Description**

Fits an ARIMA model with exogenous regressors (ARIMAX).

**Usage**

```
ts_arimax(
  x,
  xreg,
  order = NULL,
  seasonal = NULL,
  stepwise = TRUE,
  approximation = TRUE,
  ...
)
```

**Arguments**

|                                      |                                                                                                                                        |
|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| <code>x</code>                       | A complete <code>ts_df</code> with no missing values.                                                                                  |
| <code>xreg</code>                    | A matrix or <code>data.frame</code> of exogenous regressors with <code>nrow(xreg) == nrow(x)</code> .                                  |
| <code>order</code>                   | Optional non-seasonal ARIMA order. If <code>NULL</code> (default), automatic order selection via <code>forecast::auto.arima()</code> . |
| <code>seasonal</code>                | Optional seasonal specification.                                                                                                       |
| <code>stepwise, approximation</code> | Passed to <code>forecast::auto.arima()</code> .                                                                                        |
| <code>...</code>                     | Additional arguments passed to the forecast fitting function.                                                                          |

**Value**

A list containing tidy model summaries, the raw model, input, and the exogenous regressors used.

**Examples**

```
x = as_ts_df(log(AirPassengers))
xreg = data.frame(trend = seq_len(nrow(x)))
ts_arimax(x, xreg = xreg)
```

---

|                   |                                 |
|-------------------|---------------------------------|
| ts_back_transform | <i>Back-Transform Forecasts</i> |
|-------------------|---------------------------------|

---

**Description**

Converts forecasts produced on a transformed scale back to the original scale.

**Usage**

```
ts_back_transform(forecast_tbl, params)
```

**Arguments**

|              |                                                       |
|--------------|-------------------------------------------------------|
| forecast_tbl | Forecast tibble from <code>ts_forecast()</code> .     |
| params       | The params element from <code>ts_transform()</code> . |

**Value**

The forecast tibble with numeric forecast columns restored.

**Examples**

```
x = as_ts_df(AirPassengers)
tr = ts_transform(x, method = "log")
fc = tibble::tibble(step = 1:2, forecast = log(c(500, 600)))
ts_back_transform(fc, tr$params)
```

---

|       |                                           |
|-------|-------------------------------------------|
| ts_df | <i>Lightweight Time-Series Data Frame</i> |
|-------|-------------------------------------------|

---

**Description**

Creates a simple single-series time-series data frame with fixed columns time and value, plus frequency and start attributes.

**Usage**

```
ts_df(data, time = "time", value = "value", frequency = 1, start = NULL)
```

**Arguments**

|           |                                                               |
|-----------|---------------------------------------------------------------|
| data      | A data frame.                                                 |
| time      | Name of the time/index column.                                |
| value     | Name of the numeric value column.                             |
| frequency | Positive numeric scalar. Defaults to 1.                       |
| start     | Optional start metadata. If NULL, uses the first sorted time. |

**Value**

A `ts_df` object, which is also a data frame.

**Examples**

```
df = data.frame(time = c(2, 1, 3), value = c(12, 10, 15))
ts_df(df)
```

---

|        |                          |
|--------|--------------------------|
| ts_ets | <i>ETS Model Fitting</i> |
|--------|--------------------------|

---

**Description**

ETS Model Fitting

**Usage**

```
ts_ets(x, model = "ZZZ", ...)
```

**Arguments**

|       |                                                 |
|-------|-------------------------------------------------|
| x     | A complete ts_df with no missing values.        |
| model | ETS model string passed to forecast::ets().     |
| ...   | Additional arguments passed to forecast::ets(). |

**Value**

A list containing tidy model summaries, the raw model, and input.

**Examples**

```
ts_ets(as_ts_df(log(AirPassengers)))
```

---

|             |                           |
|-------------|---------------------------|
| ts_forecast | <i>Generate Forecasts</i> |
|-------------|---------------------------|

---

**Description**

Generate Forecasts

**Usage**

```
ts_forecast(model_result, h = 12, level = c(80, 95), newxreg = NULL)
```

**Arguments**

|              |                                                                                                                                                    |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| model_result | Result from ts_ets(), ts_sarima(), or ts_arimax().                                                                                                 |
| h            | Forecast horizon.                                                                                                                                  |
| level        | Confidence levels.                                                                                                                                 |
| newxreg      | Future exogenous regressors for ARIMAX models. A matrix or data.frame with nrow(newxreg) == h and the same number of columns as the original xreg. |



**Value**

A tibble with point forecasts and intervals.

**Examples**

```
fit = ts_ets(as_ts_df(log(AirPassengers)))
ts_forecast(fit, h = 3)

x = as_ts_df(log(AirPassengers))
xreg = data.frame(trend = seq_len(nrow(x)))
fit_arimax = ts_arimax(x, xreg = xreg, order = c(1, 1, 1))
newxreg = data.frame(trend = nrow(x) + 1:3)
ts_forecast(fit_arimax, h = 3, newxreg = newxreg)
```

---

|           |                             |
|-----------|-----------------------------|
| ts_sarima | <i>SARIMA Model Fitting</i> |
|-----------|-----------------------------|

---

**Description**

SARIMA Model Fitting

**Usage**

```
ts_sarima(
  x,
  order = NULL,
  seasonal = NULL,
  stepwise = TRUE,
  approximation = TRUE,
  ...
)
```

**Arguments**

|                         |                                                               |
|-------------------------|---------------------------------------------------------------|
| x                       | A complete ts_df with no missing values.                      |
| order                   | Optional non-seasonal ARIMA order.                            |
| seasonal                | Optional seasonal specification.                              |
| stepwise, approximation | Passed to forecast::auto.arima().                             |
| ...                     | Additional arguments passed to the forecast fitting function. |

**Value**

A list containing tidy model summaries, the raw model, and input.

**Examples**

```
ts_sarima(as_ts_df(log(AirPassengers)), stepwise = TRUE, approximation = TRUE)
```

---

|        |                                      |
|--------|--------------------------------------|
| ts_stl | <i>STL Decomposition for a ts_df</i> |
|--------|--------------------------------------|

---

**Description**

Decomposes a complete seasonal `ts_df` with `stats::stl()`.

**Usage**

```
ts_stl(x, s_window = "periodic", robust = TRUE, ...)
```

**Arguments**

|                       |                                                                |
|-----------------------|----------------------------------------------------------------|
| <code>x</code>        | A complete seasonal <code>ts_df</code> with no missing values. |
| <code>s_window</code> | Seasonal smoothing window.                                     |
| <code>robust</code>   | Use robust STL fitting.                                        |
| <code>...</code>      | Additional arguments passed to <code>stats::stl()</code> .     |

**Value**

A list with components `ts`, `strength`, and `model`.

**Examples**

```
ts_stl(as_ts_df(AirPassengers))
```

---

|         |                                             |
|---------|---------------------------------------------|
| ts_test | <i>Stationarity Tests for a Time Series</i> |
|---------|---------------------------------------------|

---

**Description**

Runs ADF and KPSS tests on a complete `ts_df`.

**Usage**

```
ts_test(x, adf_lags = NULL)
```

**Arguments**

|                       |                                                       |
|-----------------------|-------------------------------------------------------|
| <code>x</code>        | A complete <code>ts_df</code> with no missing values. |
| <code>adf_lags</code> | Optional lag for the ADF test.                        |

**Value**

A tibble with test results.

**Examples**

```
ts_test(as_ts_df(log(AirPassengers)))
```

---

|              |                          |
|--------------|--------------------------|
| ts_transform | <i>Transform a ts_df</i> |
|--------------|--------------------------|

---

**Description**

Applies a variance-stabilising transform and optional differencing. Returns a transformed `ts_df` plus explicit parameters for back-transformation.

**Usage**

```
ts_transform(  
  x,  
  method = c("none", "log", "boxcox"),  
  lambda = NULL,  
  diff = 0L,  
  seasonal_diff = 0L  
)
```

**Arguments**

|                            |                                                       |
|----------------------------|-------------------------------------------------------|
| <code>x</code>             | A complete <code>ts_df</code> with no missing values. |
| <code>method</code>        | One of "none", "log", or "boxcox".                    |
| <code>lambda</code>        | Optional Box-Cox lambda.                              |
| <code>diff</code>          | Number of regular differences.                        |
| <code>seasonal_diff</code> | Number of seasonal differences.                       |

**Value**

A list with transformed, params, and summary.

**Examples**

```
ts_transform(as_ts_df(AirPassengers), method = "log", diff = 1)
```

---

|                |                         |
|----------------|-------------------------|
| validate_ts_df | <i>Validate a ts_df</i> |
|----------------|-------------------------|

---

**Description**

Validate a `ts_df`

**Usage**

```
validate_ts_df(x, require_complete = TRUE, require_no_na = FALSE)
```

Arguments

- x A ts\_df object.
- require\_complete If TRUE, first and last values must not be NA.
- require\_no\_na If TRUE, no values may be NA.

Value

A sorted ts\_df object.

Examples

```
x = ts_df(data.frame(time = 1:3, value = c(10, 12, 15)))
validate_ts_df(x)
```

---

|               |                       |
|---------------|-----------------------|
| water_quality | Water Quality Dataset |
|---------------|-----------------------|

---

Description

A dataset containing water quality evaluation metrics for 20 rivers, including dissolved oxygen (O2, positive indicator), pH value (PH, centered indicator), total bacteria count (germ, negative indicator), and plant nutrient content (nutrient, interval indicator with optimal range 10-20). This dataset is suitable for multi-criteria decision analysis, such as weight calculation and fuzzy comprehensive evaluation in the mathmodels package.

Usage

```
water_quality
```

Format

- A data frame with 20 rows and 5 columns:
- ID** Numeric, unique identifier for each river (1 to 20).
  - O2** Numeric, dissolved oxygen content (mg/L), higher values are better (positive indicator).
  - PH** Numeric, pH value, values closer to 7 are optimal (centered indicator).
  - germ** Numeric, total bacteria count, lower values are better (negative indicator).
  - nutrient** Numeric, plant nutrient content (mg/L), optimal range is 10-20 (interval indicator).

Details

Water Quality Dataset

Source

Simulated data for water quality evaluation, created for demonstration purposes.

*water\_quality*

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### **Examples**

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# Load the dataset  
data(water_quality)
```

```
# Preview the data  
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```

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